Chapter 2

Motion

Introduction

An Overview of Mechanics

Mechanics: The study of how bodies react to forces acting on them.

Statics: The study of bodies in equilibrium.

Dynamics:

1. **Kinematics** – concerned with the geometric aspects of motion
2. **Kinetics** – concerned with the forces causing the motion

Introduction

- **Kinematics** is the study of motion.
- *Velocity* and *acceleration* are important physical quantities.
- A typical runner gains speed gradually during the course of a sprinting foot race and then slows down after crossing the finish line.

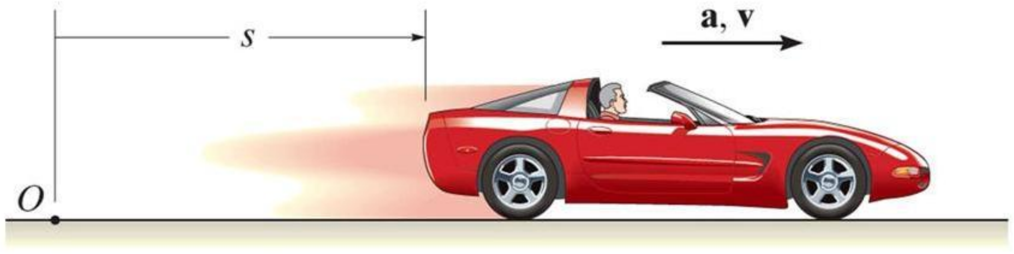
Introduction

The motion of large objects, such as rockets, airplanes, or cars, can often be analyzed as if they were particles.

Why?

If we measure the altitude of this rocket as a function of time, how can we determine its velocity and acceleration?

Introduction



A sports car travels along a straight road.

Can we treat the car as a particle?

If the car accelerates at a constant rate, how can we determine its position and velocity at some instant?

Displacement, time, and average velocity

- A particle moving along the x-axis has a coordinate x.
- The change in the particle's coordinate is $$\Delta x = x_2 - x_1.$$
- The average x-velocity of the particle is $$v_{\text{av-}x} = \Delta x / \Delta t.$$

Average x-velocity of a particle in **straight-line motion** during time interval from t_1 to t_2

x-component of the particle's displacement

$$v_{\text{av-}x} = \frac{\Delta x}{\Delta t} = \frac{x_2 - x_1}{t_2 - t_1}$$

Final x-coordinate minus initial x-coordinate

Time interval

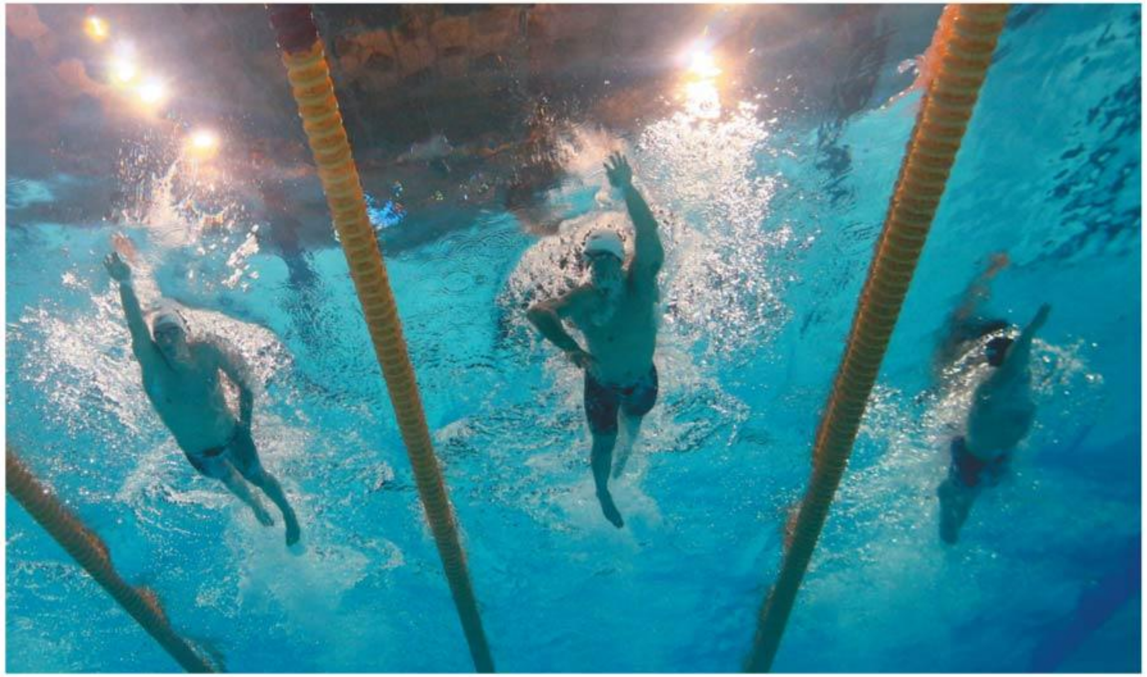
Final time minus initial time

Rules for the sign of *x*-velocity

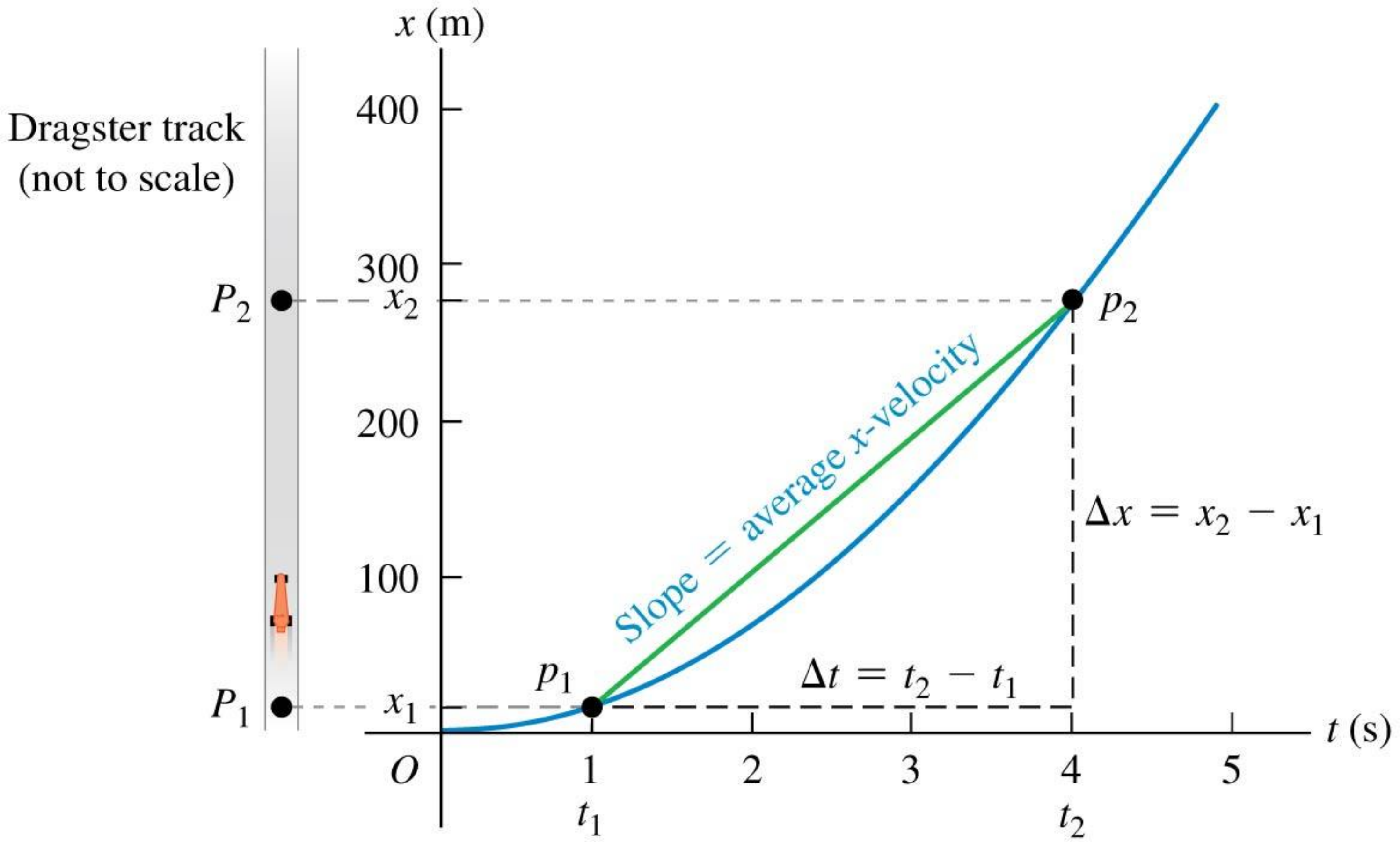
If x-coordinate is:	. . . x-velocity is:
Positive & increasing (getting more positive)	Positive: Particle is moving in $+x$-direction
Positive & decreasing (getting less positive)	Negative: Particle is moving in $-x$-direction
Negative & increasing (getting less negative)	Positive: Particle is moving in $+x$-direction
Negative & decreasing (getting more negative)	Negative: Particle is moving in $-x$-direction

Average velocity

- The winner of a 50-m swimming race is the swimmer whose average velocity has the greatest magnitude.
- That is, the swimmer who traverses a displacement Δx of 50 m in the shortest elapsed time Δt.



A position-time graph



Instantaneous velocity

- The **instantaneous velocity** is the velocity at a specific instant of time or specific point along the path and is given by $v_x = dx/dt$.
- The average speed is *not* the magnitude of the average velocity!

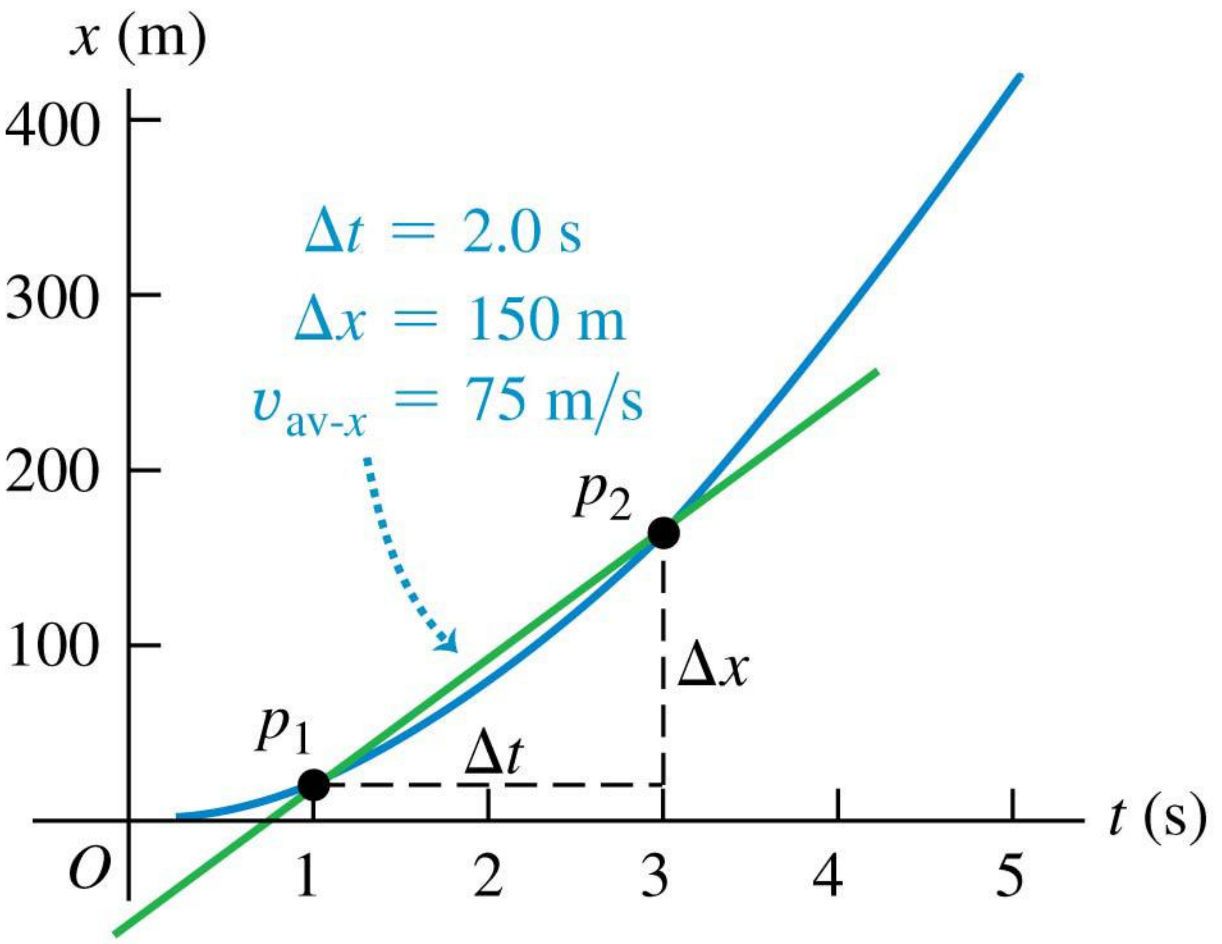
The **instantaneous *x*-velocity** of a particle in **straight-line motion** ...

$$v_x = \lim_{\Delta t \to 0} \frac{\Delta x}{\Delta t} = \frac{dx}{dt}$$

... equals the limit of the particle's average *x*-velocity as the time interval approaches zero ...

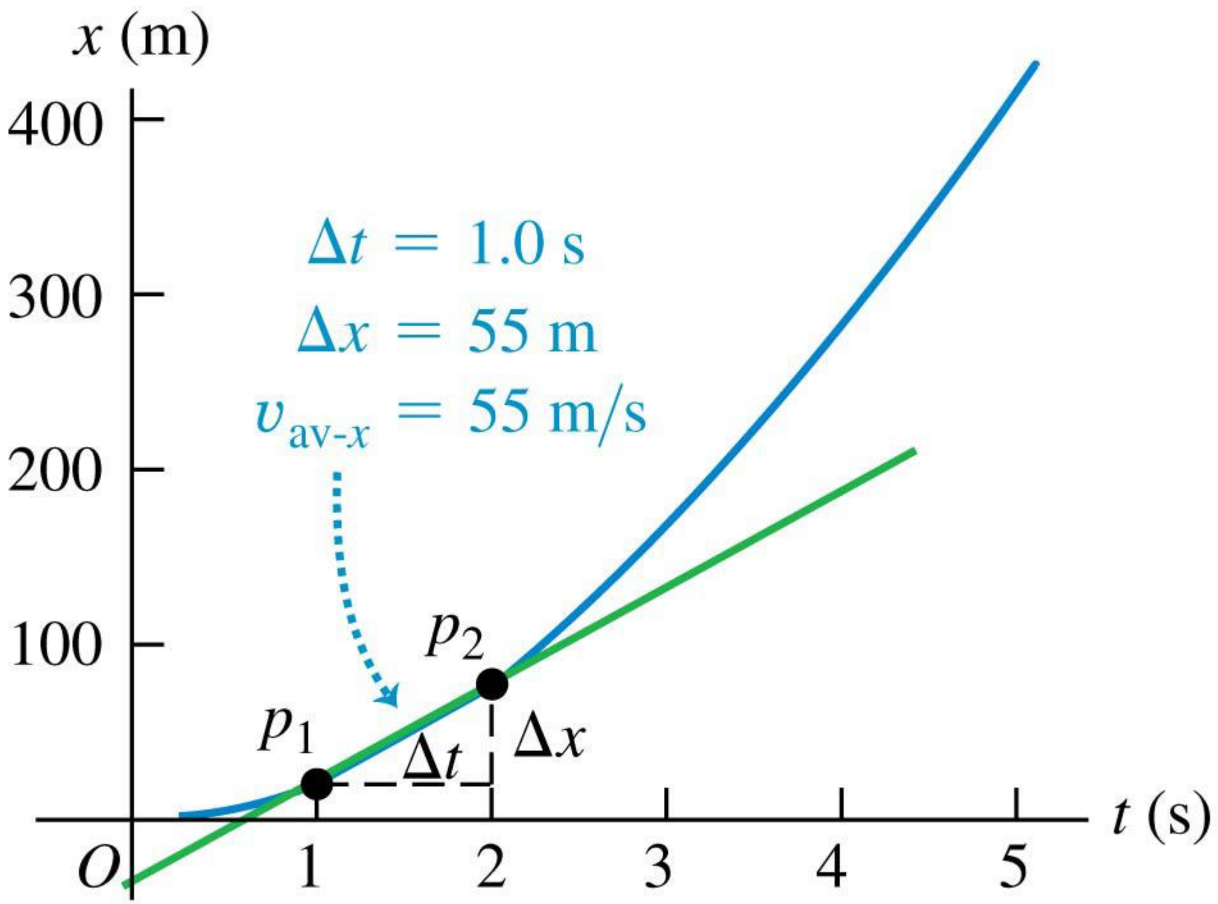
... and equals the instantaneous rate of change of the particle's *x*-coordinate.

Finding velocity on an *x*-*t* graph



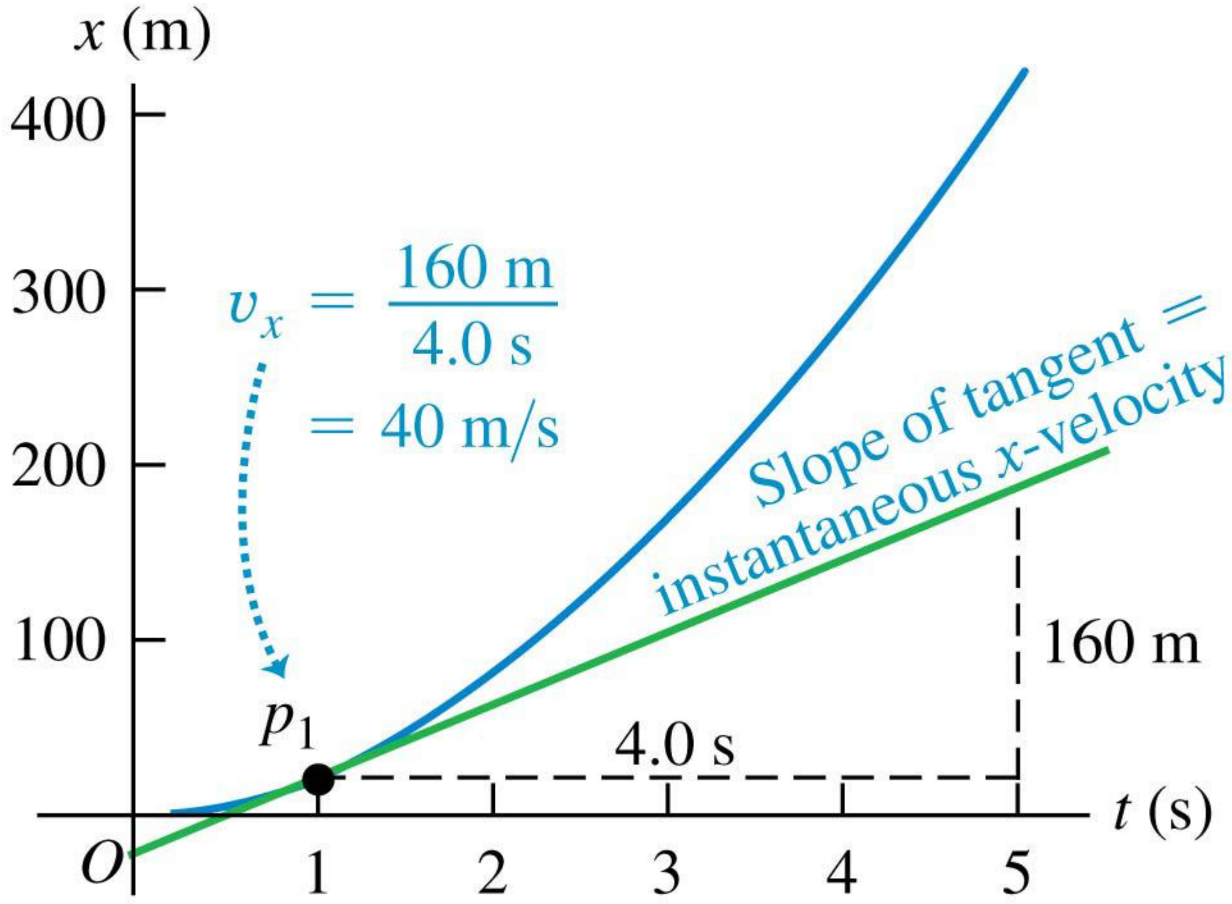
As the average x-velocity $v_{av\text{-}x}$ is calculated over shorter and shorter time intervals ...

Finding velocity on an *x-t* graph



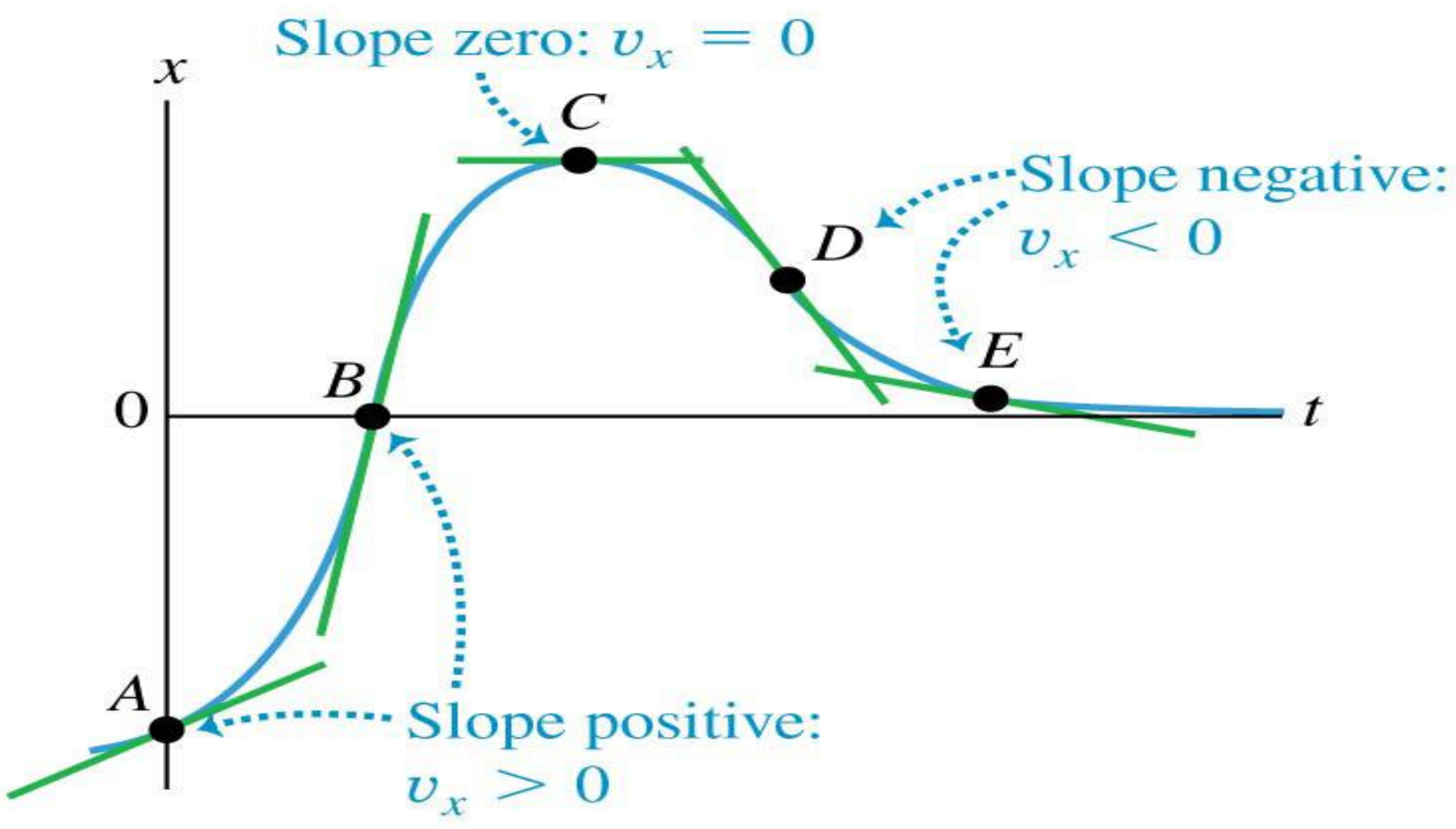
... its value $v_{\text{av-}x} = \Delta x/\Delta t$ approaches the instantaneous x-velocity.

Finding velocity on an *x*-*t* graph



The instantaneous *x*-velocity v_x at any given point equals the slope of the tangent to the *x*-*t* curve at that point.

x-t graphs



Slope zero: $v_x = 0$
x
C
Slope negative:
$v_x < 0$
D
E
B
0
t
A
Slope positive:
$v_x > 0$

Average acceleration

- Acceleration describes the rate of change of velocity with time.

- The *average x*-acceleration is

$$a_{\text{av-}x} = \Delta v_x / \Delta t.$$

Average *x*-acceleration of a particle in **straight-line motion** during time interval from t_1 to t_2

Change in *x*-component of the particle's velocity

$$a_{\text{av-}x} = \frac{\Delta v_x}{\Delta t} = \frac{v_{2x} - v_{1x}}{t_2 - t_1}$$

Final *x*-velocity minus initial *x*-velocity

Time interval

Final time minus initial time

Instantaneous acceleration

- The *instantaneous* acceleration is $a_x = dv_x/dt$.

The **instantaneous** **x-acceleration** of a particle in **straight-line motion** ...

$$a_x = \lim_{\Delta t \to 0} \frac{\Delta v_x}{\Delta t} = \frac{dv_x}{dt}$$

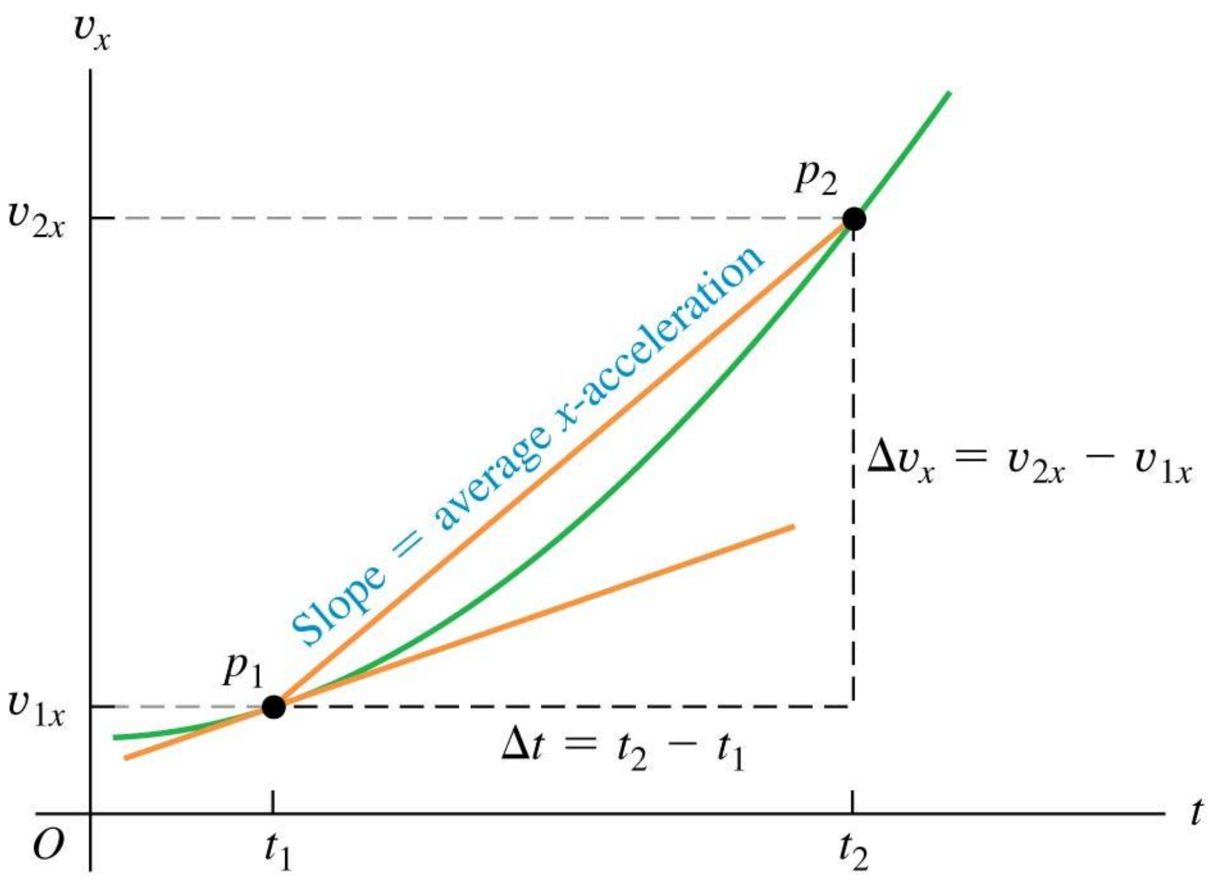
... equals the limit of the particle's average x-acceleration as the time interval approaches zero ...

... and equals the instantaneous rate of change of the particle's x-velocity.

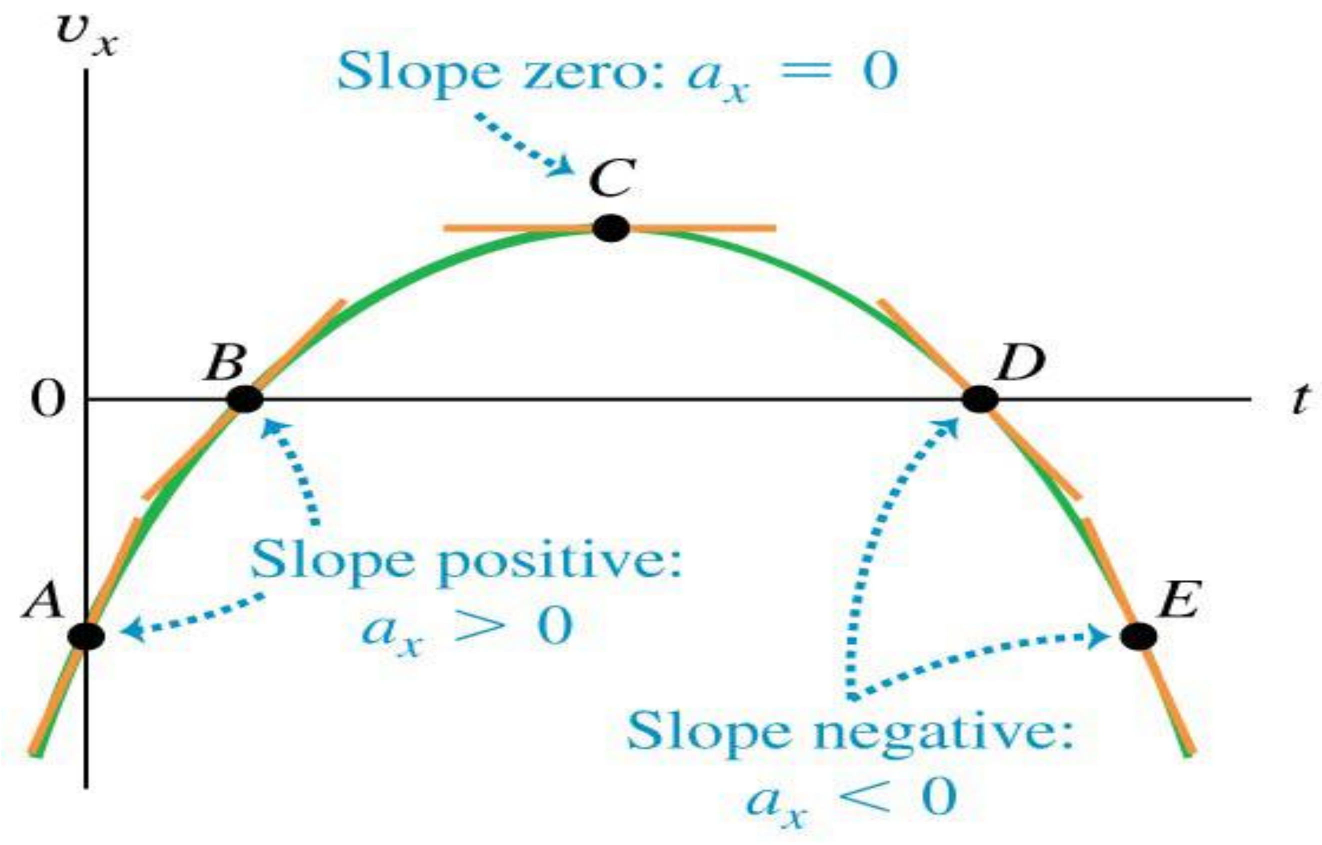
Rules for the sign of *x*-acceleration

If x-velocity is:	. . . x-acceleration is:
Positive & increasing (getting more positive)	Positive: Particle is moving in $+x$-direction & speeding up
Positive & decreasing (getting less positive)	Negative: Particle is moving in $+x$-direction & slowing down
Negative & increasing (getting less negative)	Positive: Particle is moving in $-x$-direction & slowing down
Negative & decreasing (getting more negative)	Negative: Particle is moving in $-x$-direction & speeding up

Finding acceleration on a v_x-t graph

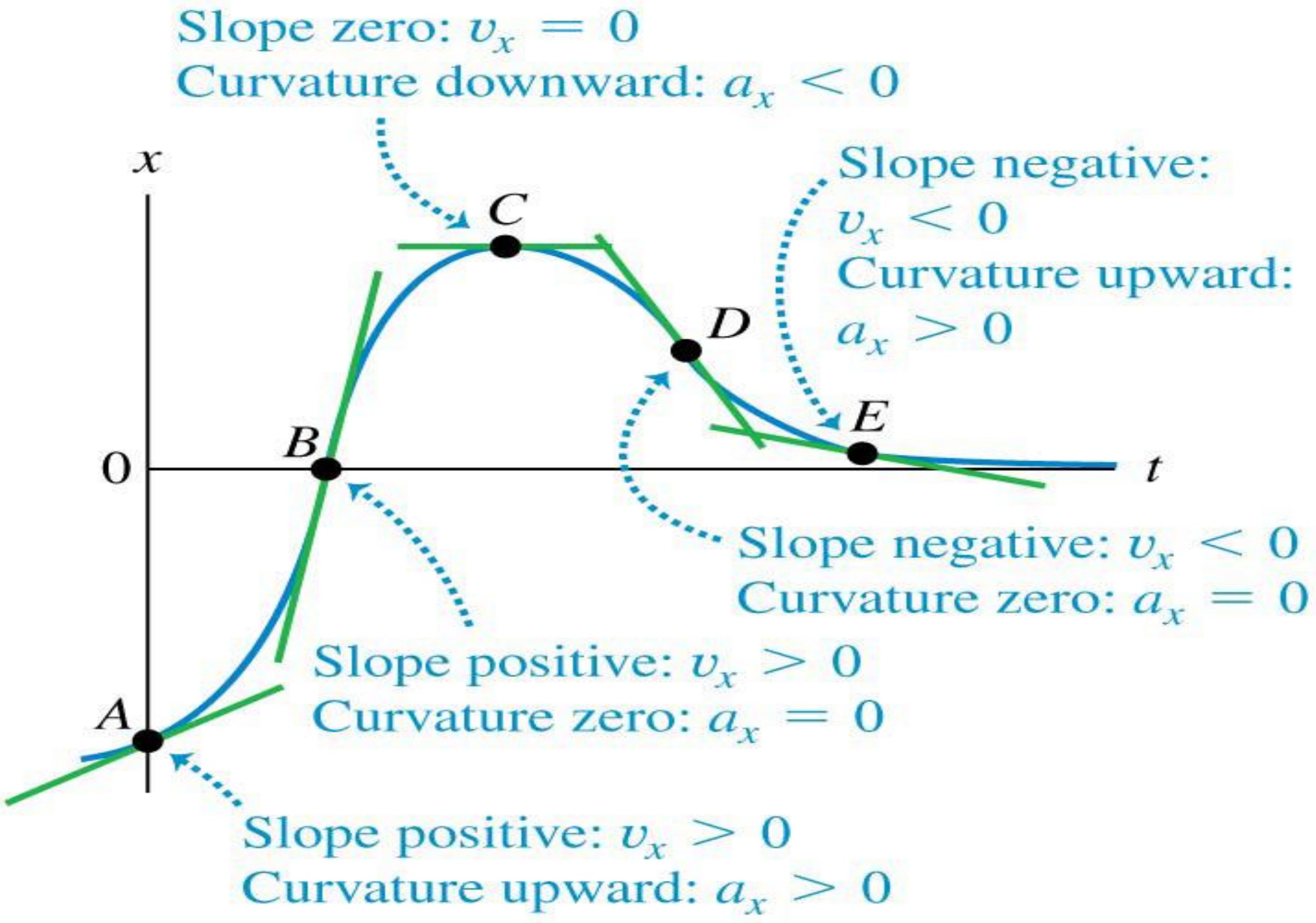


A v_x-t graph



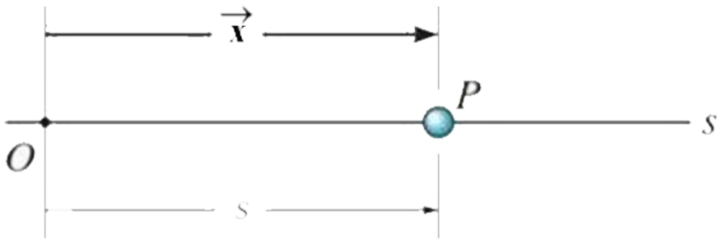
v_x
Slope zero: $a_x = 0$
C
B
D
0
t
A
E
Slope positive:
$a_x > 0$
Slope negative:
$a_x < 0$

A *x*-*t* graph

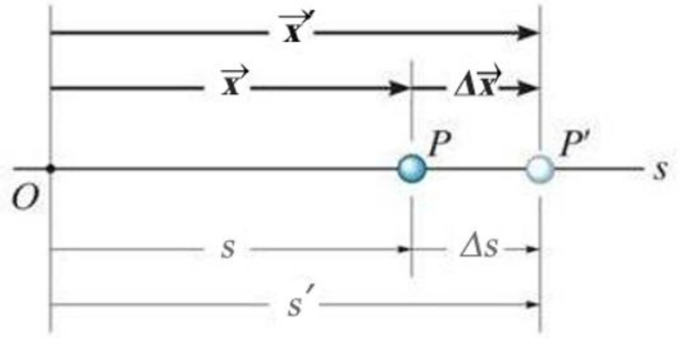


Slope zero: $v_x = 0$
Curvature downward: $a_x < 0$
x
C
D
Slope negative:
$v_x < 0$
Curvature upward:
$a_x > 0$
E
B
0
t
Slope negative: $v_x < 0$
Curvature zero: $a_x = 0$
Slope positive: $v_x > 0$
Curvature zero: $a_x = 0$
A
Slope positive: $v_x > 0$
Curvature upward: $a_x > 0$

Rectilinear Kinematics: General Formalism



Position



Displacement

A particle travels along a straight-line path defined by the coordinate axis s.

The position of the particle at any instant, relative to the origin, O, is defined by the position vector $\vec{x}$, or the scalar s. Scalar s can be positive or negative. Typical units for $\vec{x}$ and s are meters (m) or feet (ft).

The displacement of the particle is defined as its change in position.

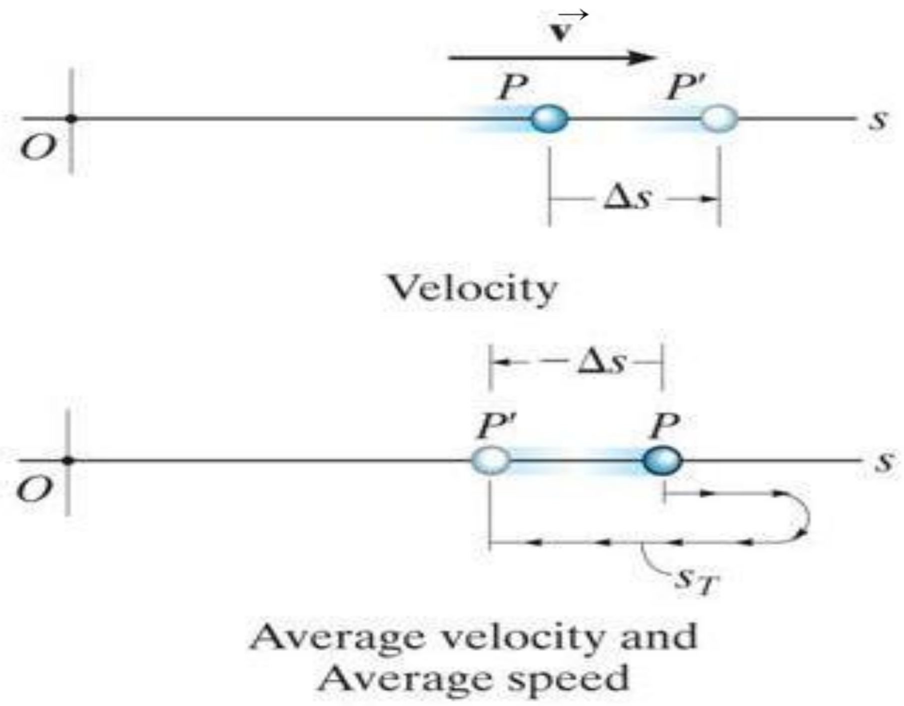
Vector form: $\Delta \vec{x} = \vec{x}' - \vec{x}$

Scalar form: $\Delta s = s' - s$

The total distance traveled by the particle, s_T, is a positive scalar that represents the total length of the path over which the particle travels.

Rectilinear Kinematics: General Formalism

Velocity is a measure of the rate of change in the position of a particle. It is a vector quantity (it has both magnitude and direction). The magnitude of the velocity is called speed, with units of m/s or ft/s.



The average velocity of a particle during a time interval Δt is

$$\vec{v}_{avg} = \Delta \vec{x} \,/\, \Delta t$$

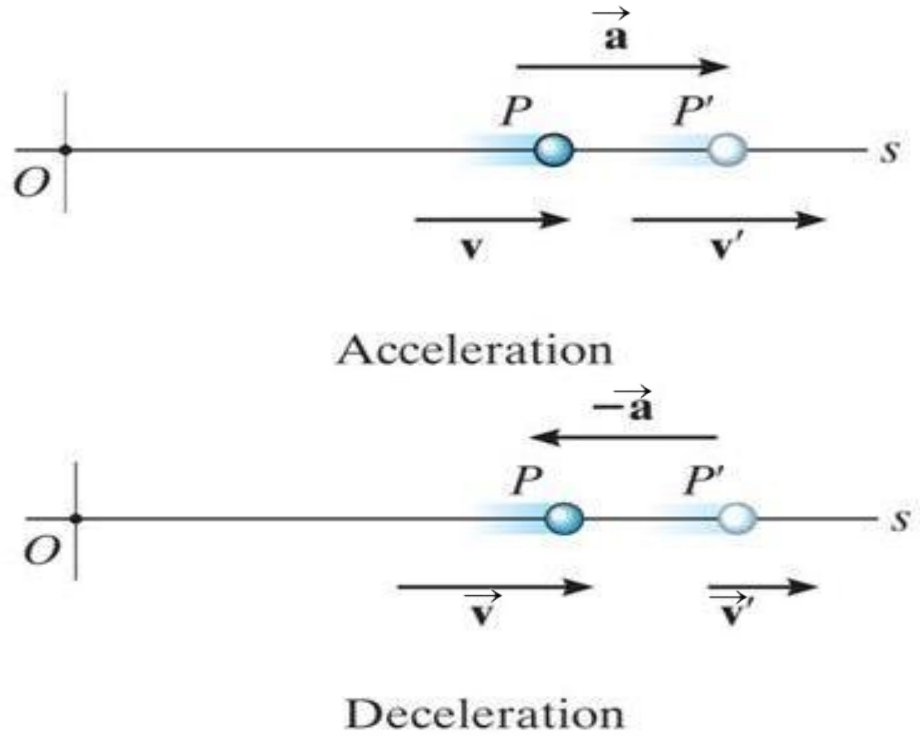
The instantaneous velocity is the time-derivative of position. $\vec{v} = d\,\vec{x} \,/\, dt$

Speed is the magnitude of velocity:
$v = ds / dt$

Average speed is the total distance traveled divided by elapsed time: $(v_{sp})_{avg} = s_T / \Delta t$

Rectilinear Kinematics: General Formalism

Acceleration is the rate of change in the velocity of a particle. It is a vector quantity. Typical units are m/s^2.



The instantaneous acceleration is the time derivative of velocity.

Vector form: $\vec{a} = d\,\vec{v} \,/\, dt$

Scalar form: $a = dv / dt = d^2s / dt^2$

Acceleration can be positive (speed increasing) or negative (speed decreasing).

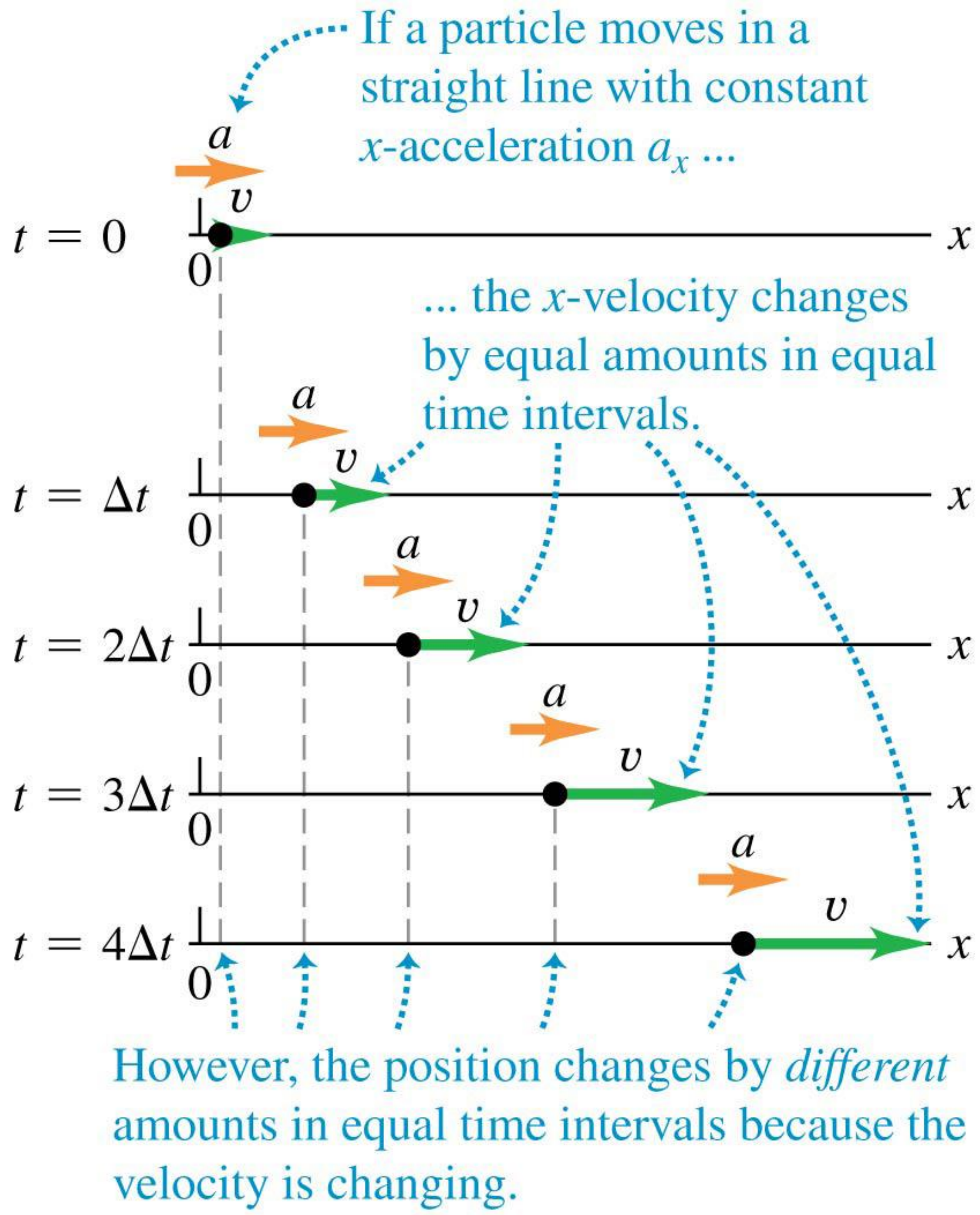
The derivative equations for velocity and acceleration can be manipulated to get $a\,ds = v\,dv$

Quiz

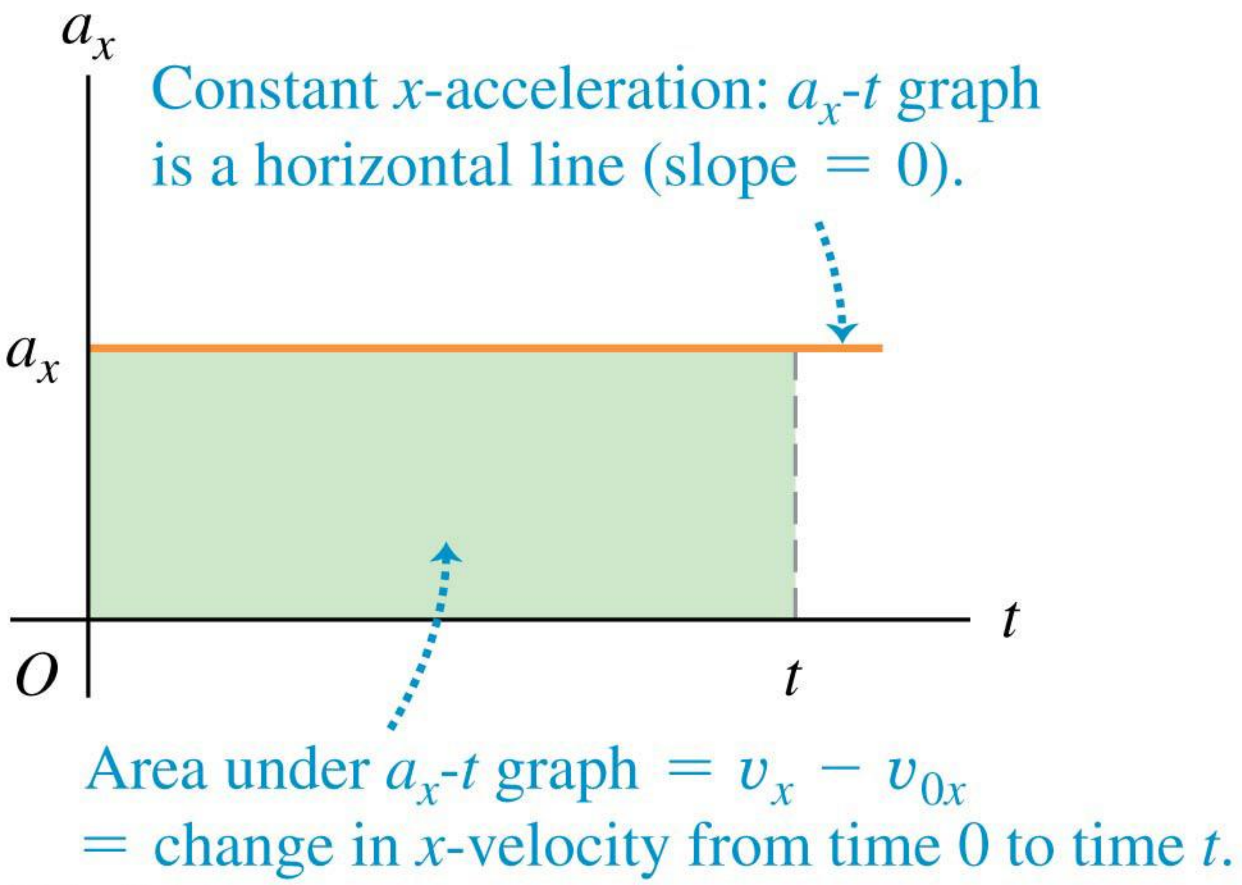
1. In dynamics, a particle is assumed to have __________.
   A) both translational and rotational motions
   B) only a mass
   C) a mass but the size and shape cannot be neglected
   D) no mass or size or shape, it is just a point

2. The average speed is defined as __________.
   A) $\Delta x / \Delta t$ B) $\Delta s / \Delta t$
   C) $s_T / \Delta t$ D) None of the above.

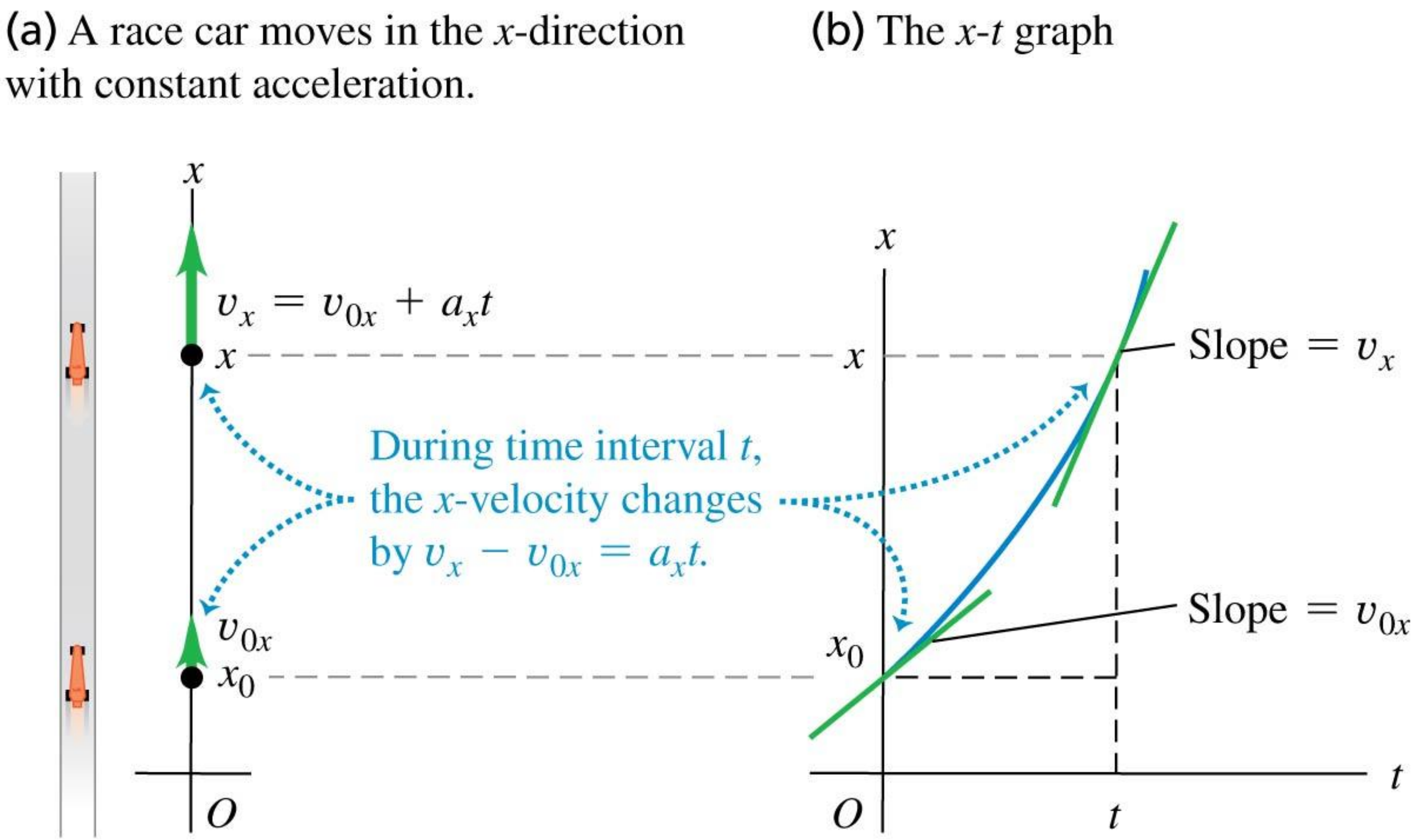
Motion with constant acceleration



Motion with constant acceleration



A position-time graph



The equations of motion with constant acceleration

- The four equations apply to any straight-line motion with constant acceleration a_x.
- First equation gives the velocity at time t
- Second equation gives the position of the particle at time t

$$v_x = v_{0x} + a_x t$$

$$x = x_0 + v_{0x} t + \frac{1}{2} a_x t^2$$

$$v_x^2 = v_{0x}^2 + 2a_x \left(x - x_0 \right)$$

$$x - x_0 = \left(\frac{v_{0x} + v_x}{2} \right) t$$

You should be able to solve most of the problems using these four equations

The equations of motion with constant acceleration

- Third equation gives you the velocity at t in terms of the initial velocity, acceleration and the distance travelled (displacement between t=0 and t).
- Fourth equation gives the distance travelled in terms of initial velocity, final velocity and time spent.

$$v_x = v_{0x} + a_x t$$

$$x = x_0 + v_{0x} t + \frac{1}{2} a_x t^2$$

$$v_x^2 = v_{0x}^2 + 2a_x \left(x - x_0 \right)$$

$$x - x_0 = \left(\frac{v_{0x} + v_x}{2} \right) t$$

Derivations

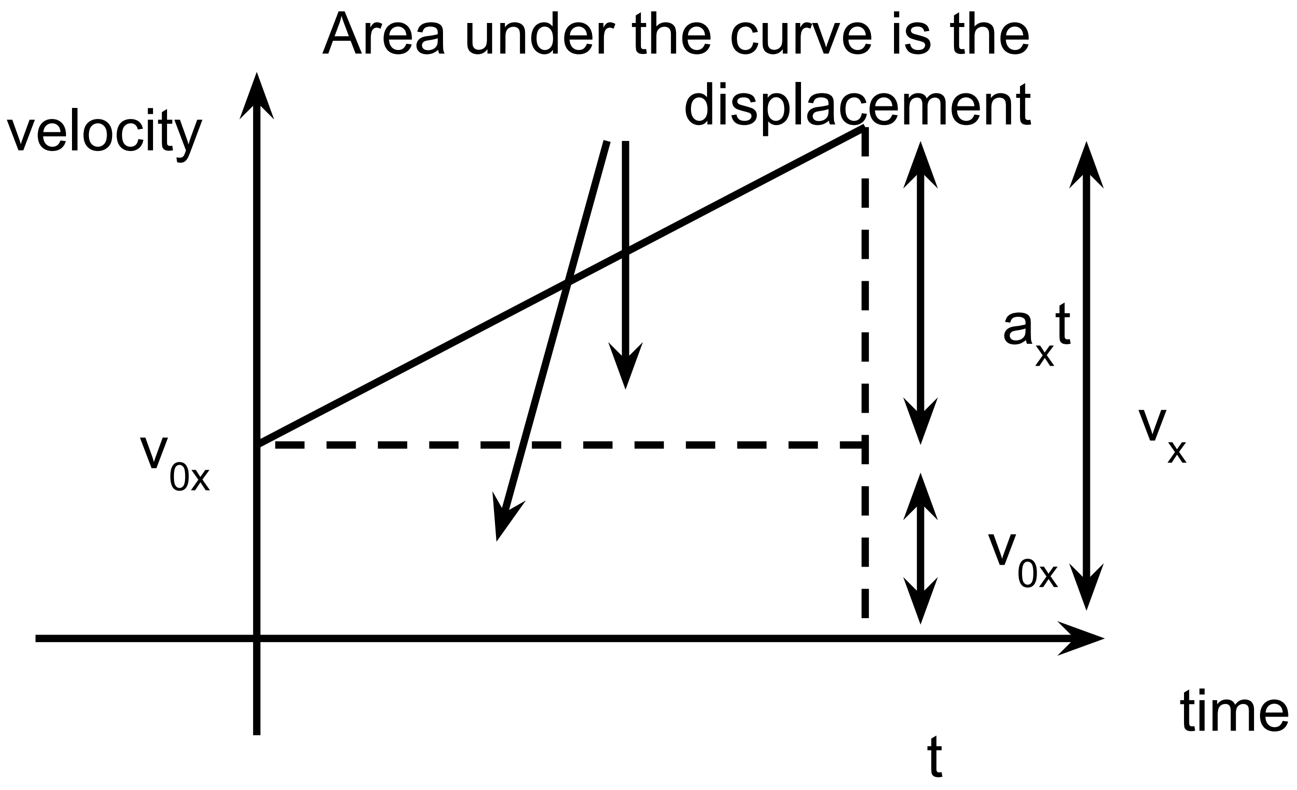
- The 1st ,2nd and 4th equations can be derived using the velocity and time graph.
- 3rd equation derived from 1st and 2nd

$$v_x = v_{0x} + a_x t$$

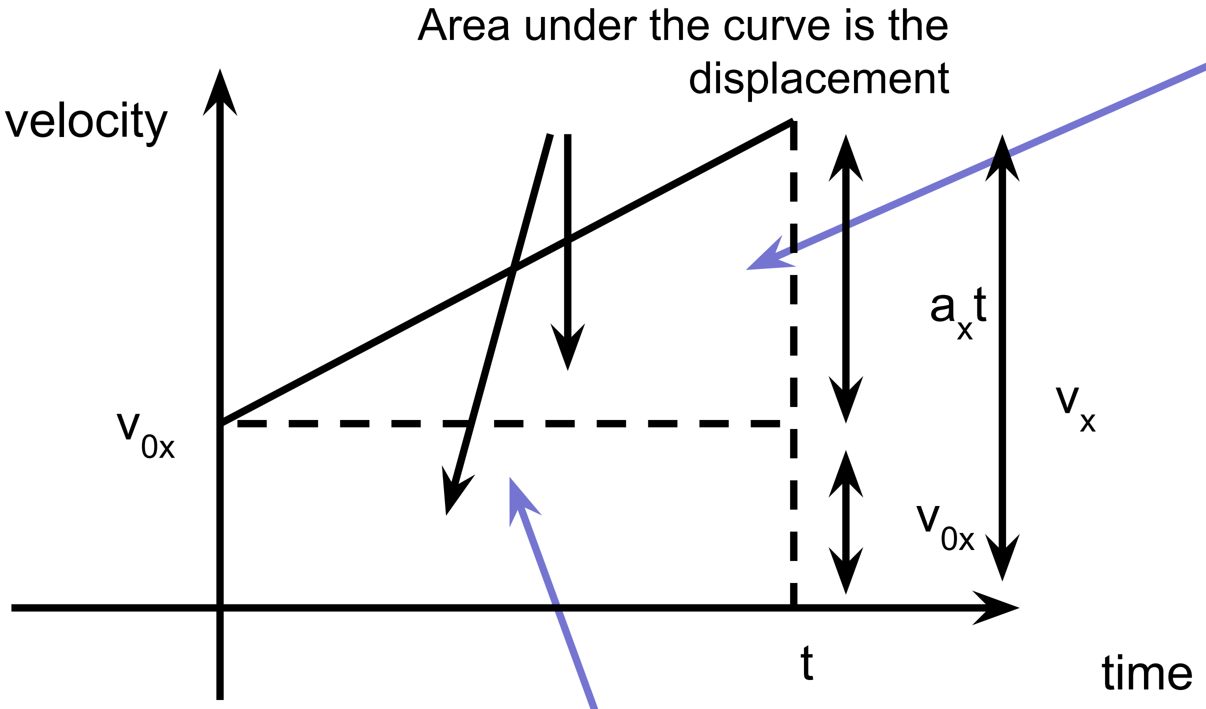
$$x = x_0 + v_{0x} t + \frac{1}{2} a_x t^2$$

$$v_x^2 = v_{0x}^2 + 2a_x \left(x - x_0 \right)$$

$$x - x_0 = \left(\frac{v_{0x} + v_x}{2} \right) t$$



Derivations



1st eqt.

$a_x t = v_x - v_{0x}$

$v_x = v_{0x} + a_x t$

2nd & 4th eqt.

Area$=x-x_0=(v_x+v_{0x})t/2$

$x-x_0=(v_{0x}+v_{0x}+a_x t)t/2=v_{0x}t+a_x t^2/2$

3rd eqt.

$$v_x = v_{0x} + a_x t \Rightarrow v_{0x} = v_x - a_x t$$

$$\Rightarrow v_{0x}^2 = v_x^2 - 2a_x v_x t + a_x^2 t^2$$

$$x = x_0 + v_{0x}t + \frac{1}{2}a_x t^2$$

$$\Rightarrow x - x_0 = v_{0x}t + \frac{1}{2}a_x t^2$$

$$\Rightarrow x - x_0 = (v_x - a_x t)t + \frac{1}{2}a_x t^2$$

$$\Rightarrow 2a_x(x - x_0) = 2a_x v_x t - a_x^2 t^2$$

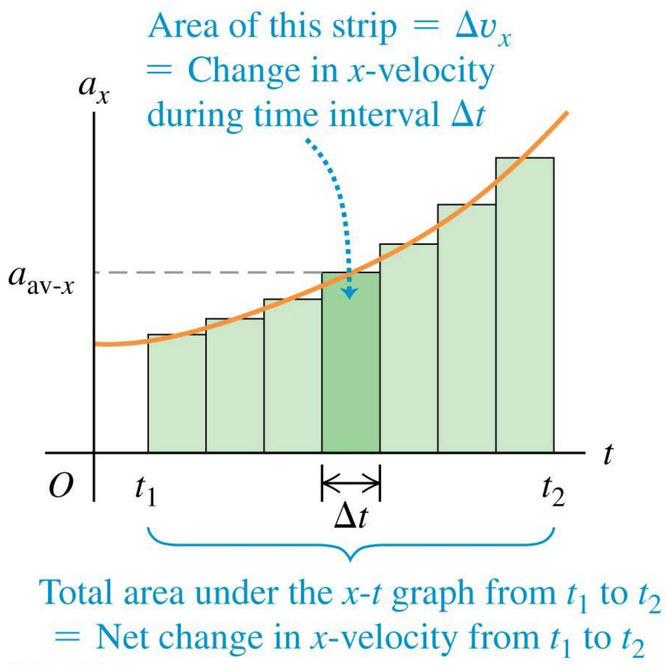
$$\Rightarrow 2a_x(x - x_0) = v_x^2 - v_{0x}^2$$

$$\Rightarrow v_x^2 = v_{0x}^2 + 2a_x(x - x_0)$$

Velocity and position by integration

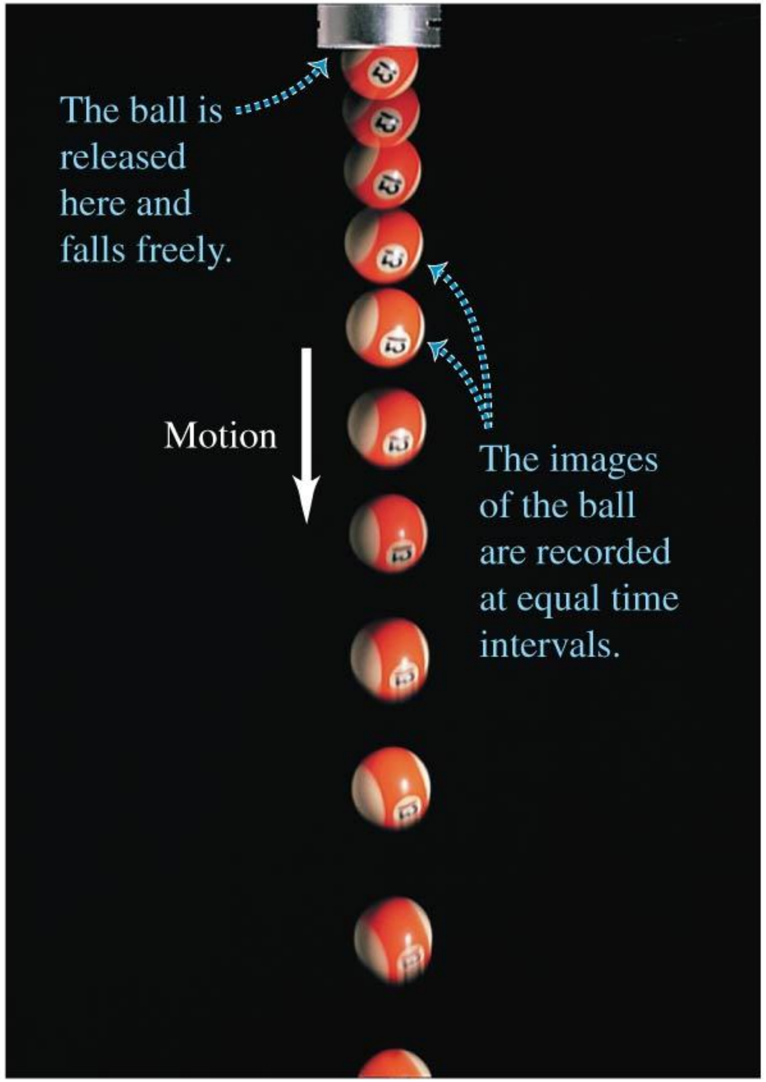
- The acceleration of a car is not always constant.
- The motion may be integrated over many small time intervals to give

$$v_x = v_{0x} + \int_0^t a_x dt \qquad x = x_0 + \int_0^t v_x dt.$$



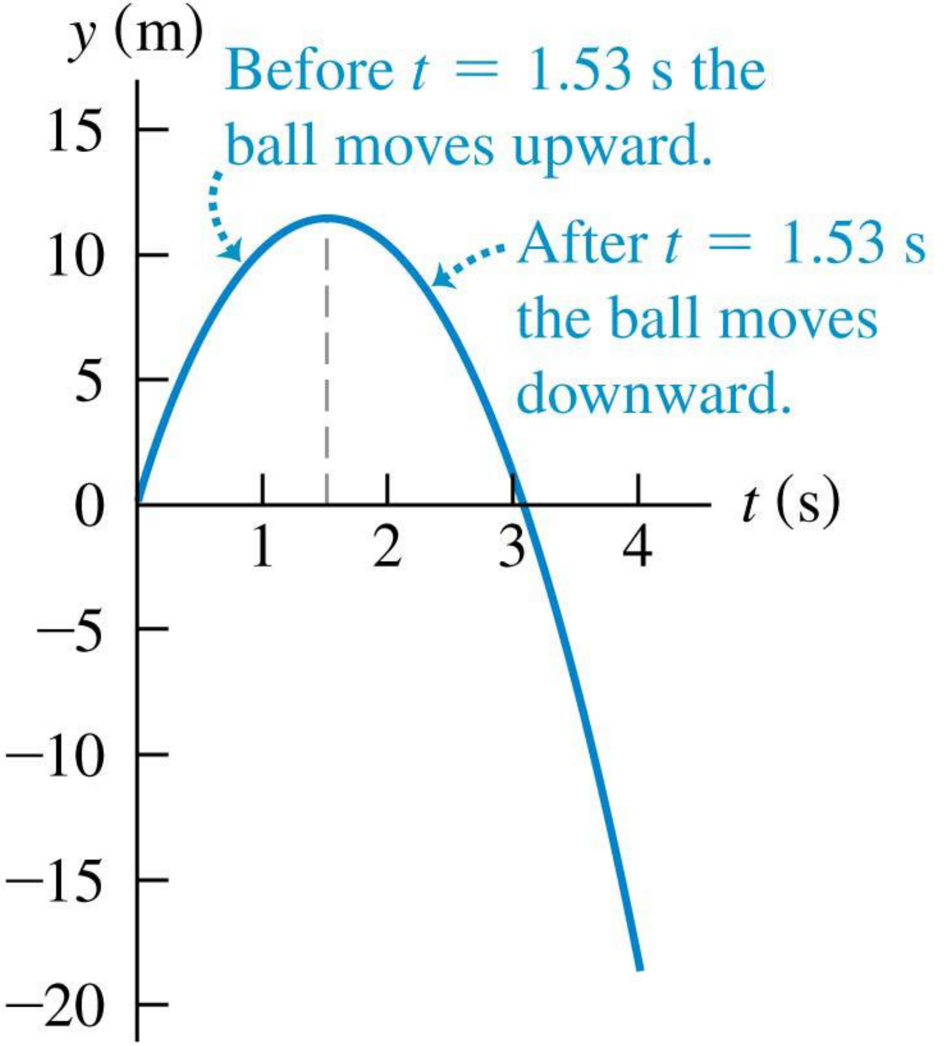
Freely falling bodies

- **Free fall** is the motion of an object under the influence of only gravity.
- In the figure, a strobe light flashes with equal time intervals between flashes.
- The velocity change is the same in each time interval, so the acceleration is constant $g = 9.8\ \text{m/s}^2$



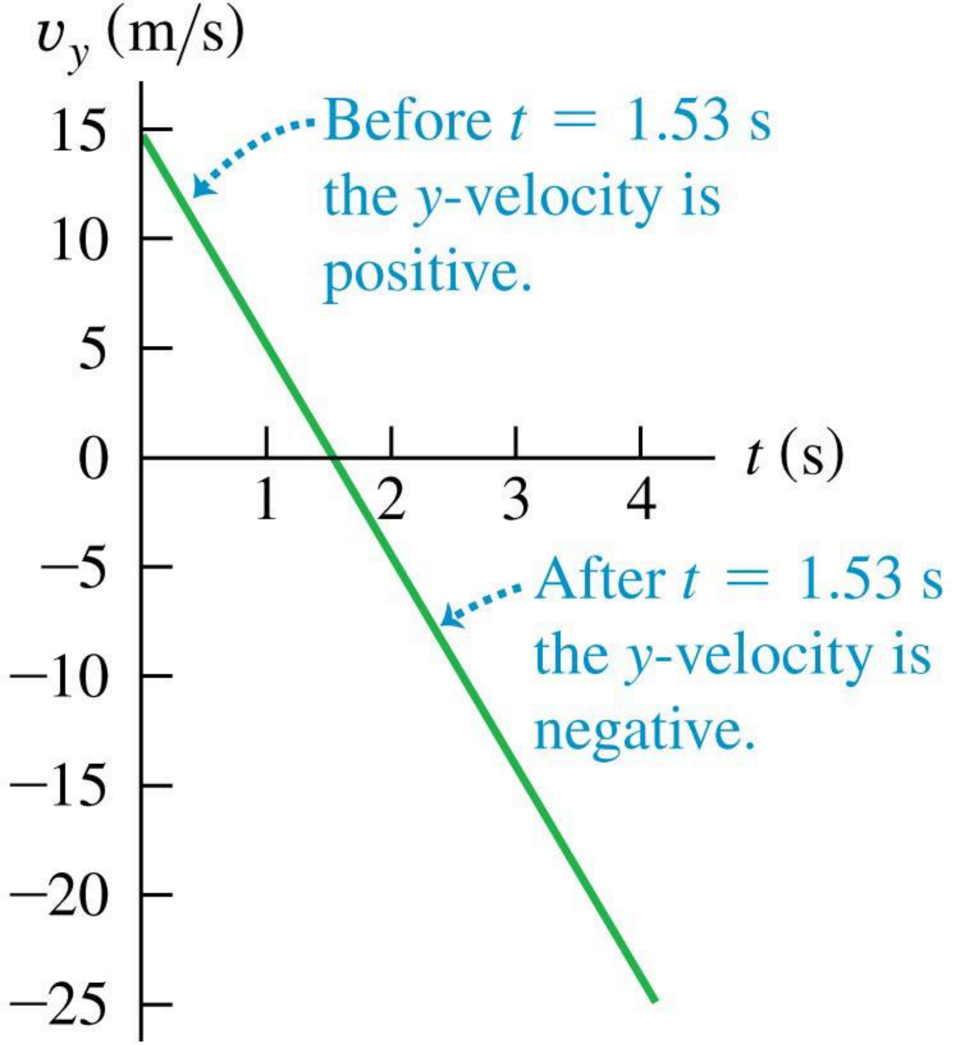
Up-and-down motion in free fall

- Position as a function of time for a ball thrown upward with an initial speed of 15.0 m/s.



Up-and-down motion in free fall

- Velocity as a function of time for a ball thrown upward with an initial speed of 15.0 m/s.
- The vertical velocity, but *not the acceleration*, is zero at the highest point.



Rectilinear kinematics: general formalism

- Differentiate position to get velocity and acceleration.

$$v = dx/dt \; ; \qquad a = dv/dt \quad \text{or} \quad a = v\, dv/dx$$

- Integrate acceleration for velocity and position.

Velocity:

$$\int_{v_o}^{v} dv = \int_{o}^{t} \mathrm{a}\, dt \quad \text{or} \quad \int_{v_o}^{v} v\, dv = \int_{x_o}^{x} \mathrm{a}\, dx$$

Position:

$$\int_{x_o}^{x} dx = \int_{o}^{t} v\, dt$$

- Note that x_o and v_o represent the initial position and velocity of the particle at $t = 0$.

Example

Given: A particle is moving along a straight line such that its velocity is defined as $v = (-4s^2)$ m/s, where s is in meters.

Find: The velocity and acceleration as functions of time if $s = 2$ m when $t = 0$.

Quiz

1. A particle has an initial velocity of 3 m/s to the left at $s_0 = 0$ m. Determine its position when $t = 3$ s if the acceleration is 2 m/s^2 to the right.

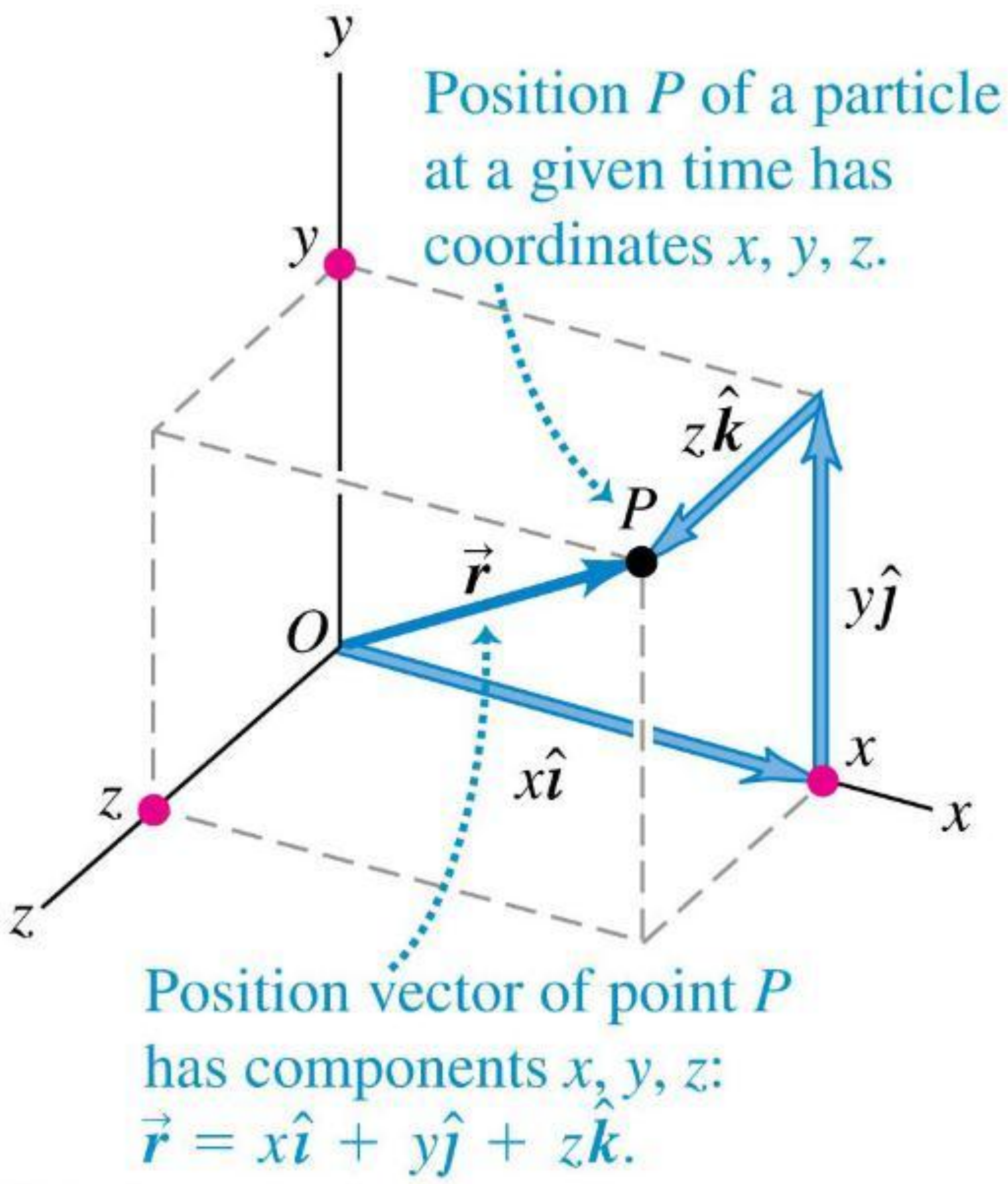
   A) 0.0 m B) 6.0 m
   C) 18.0 m D) 9.0 m

2. A particle is moving with an initial velocity of $v = 12$ m/s and constant acceleration of 3.78 m/s^2 in the same direction as the velocity. Determine the distance the particle has traveled when the velocity reaches 30 m/s.

   A) 50 m B) 100 m
   C) 150 m D) 200 m

Position vector in 3D

- The position vector from the origin to point P has components x, y, and z.



Velocity

- We define the **average velocity** as the displacement divided by the time interval:

Change in the particle's position vector

Average velocity vector of a particle during time interval from t_1 to t_2

$$\vec{v}_{av} = \frac{\Delta\vec{r}}{\Delta t} = \frac{\vec{r}_2 - \vec{r}_1}{t_2 - t_1}$$

Final position minus initial position

Time interval

Final time minus initial time

- **Instantaneous velocity** (a.k.a. "velocity") is the instantaneous rate of change of position with time:

The **instantaneous velocity vector** of a particle ...

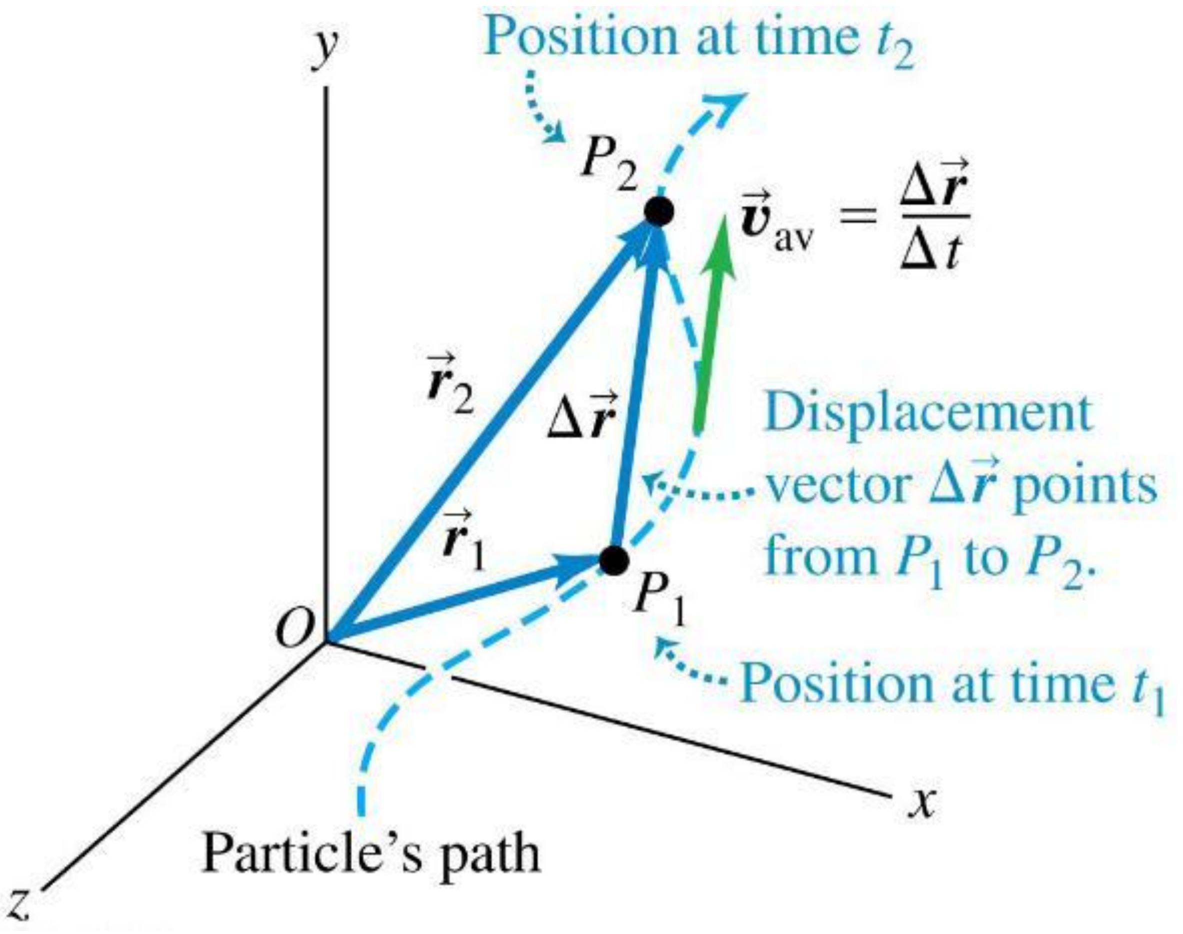
$$\vec{v} = \lim_{\Delta t \to 0} \frac{\Delta\vec{r}}{\Delta t} = \frac{d\vec{r}}{dt}$$

... equals the limit of its average velocity vector as the time interval approaches zero ...

... and equals the instantaneous rate of change of its position vector.

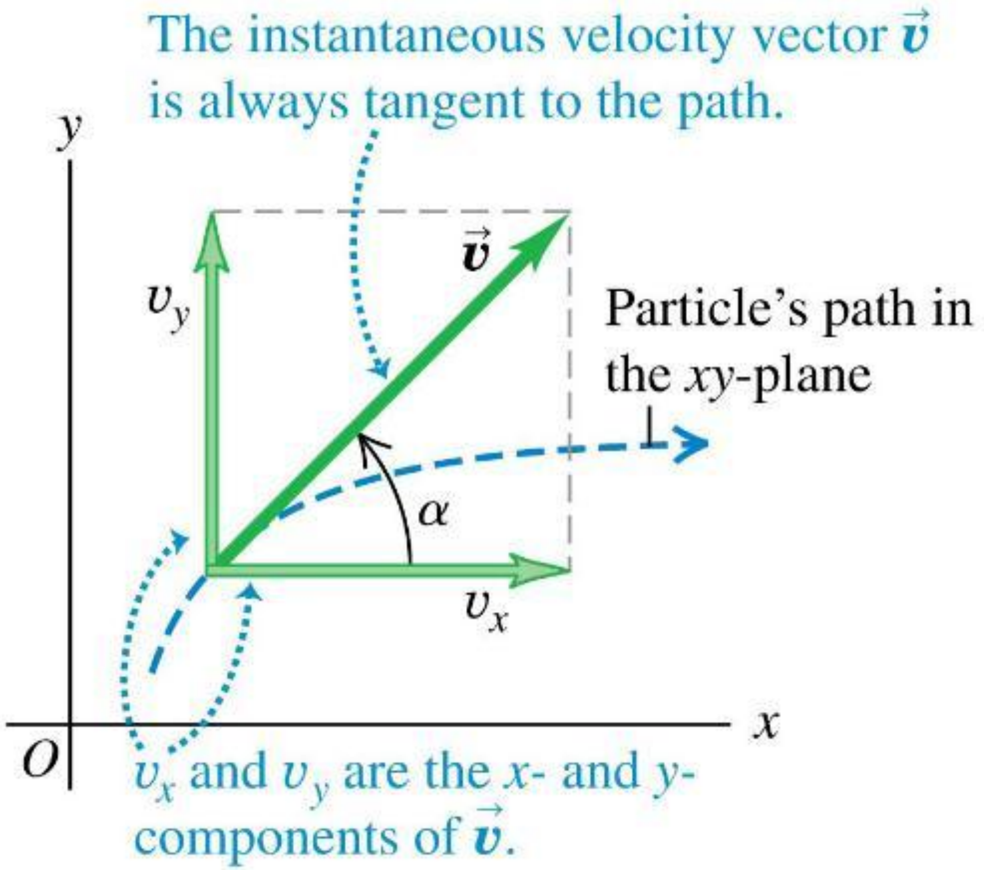
Average velocity

- The **average velocity** between two points is the displacement divided by the time interval between the two points, and it has the same direction as the displacement.



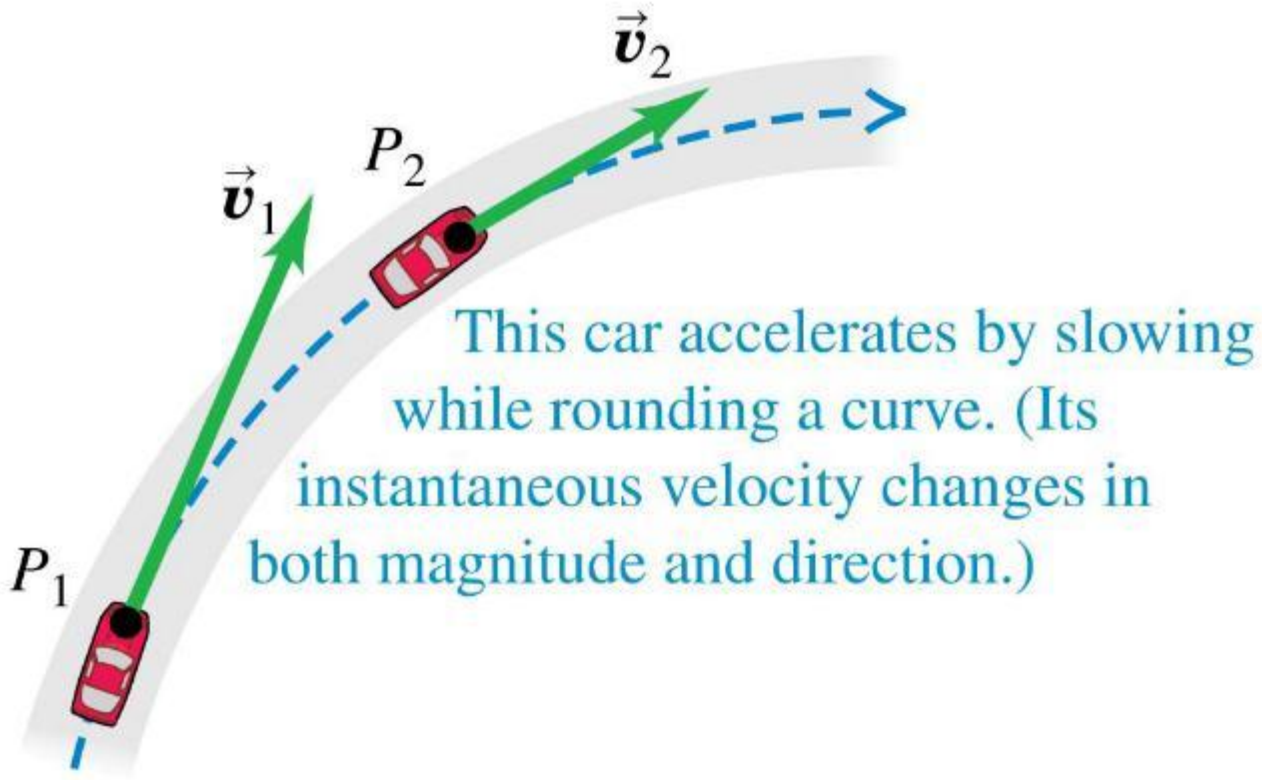
Instantaneous velocity

- The **instantaneous velocity** is the instantaneous rate of change of position vector with respect to time.
- The components of the instantaneous velocity are $v_x = dx/dt$, $v_y = dy/dt$, and $v_z = dz/dt$.
- The instantaneous velocity of a particle is always tangent to its path.



Acceleration

- Acceleration describes how the velocity changes.



Acceleration

- We define the **average acceleration** as the change in velocity divided by the time interval:

Change in the particle's velocity

Average acceleration vector of a particle during time interval from t_1 to t_2

$$\vec{a}_{av} = \frac{\Delta\vec{v}}{\Delta t} = \frac{\vec{v}_2 - \vec{v}_1}{t_2 - t_1}$$

Final velocity minus initial velocity

Time interval

Final time minus initial time

- **Instantaneous acceleration** (a.k.a. "acceleration") is the instantaneous rate of change of velocity with time:

The **instantaneous acceleration vector** of a particle ...

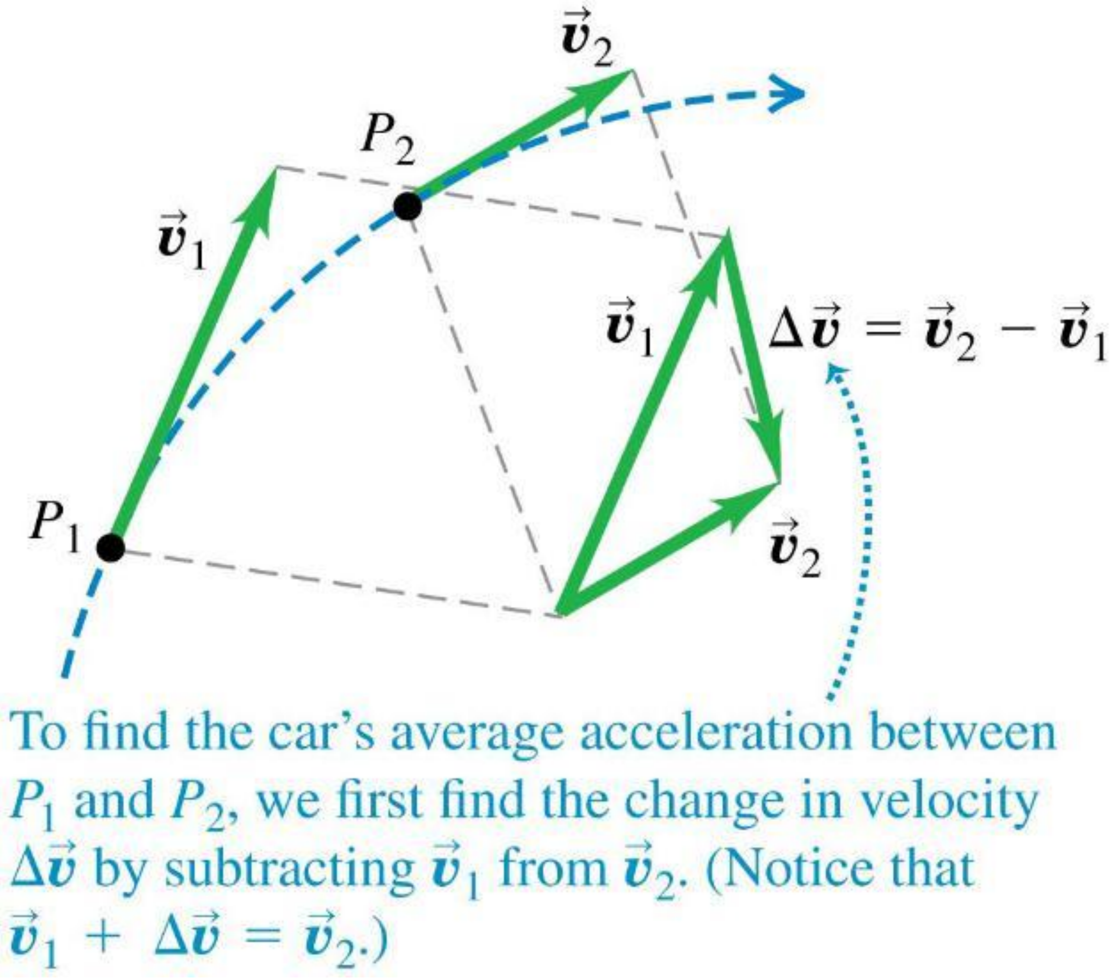
$$\vec{a} = \lim_{\Delta t \to 0} \frac{\Delta\vec{v}}{\Delta t} = \frac{d\vec{v}}{dt}$$

... equals the limit of its average acceleration vector as the time interval approaches zero ...

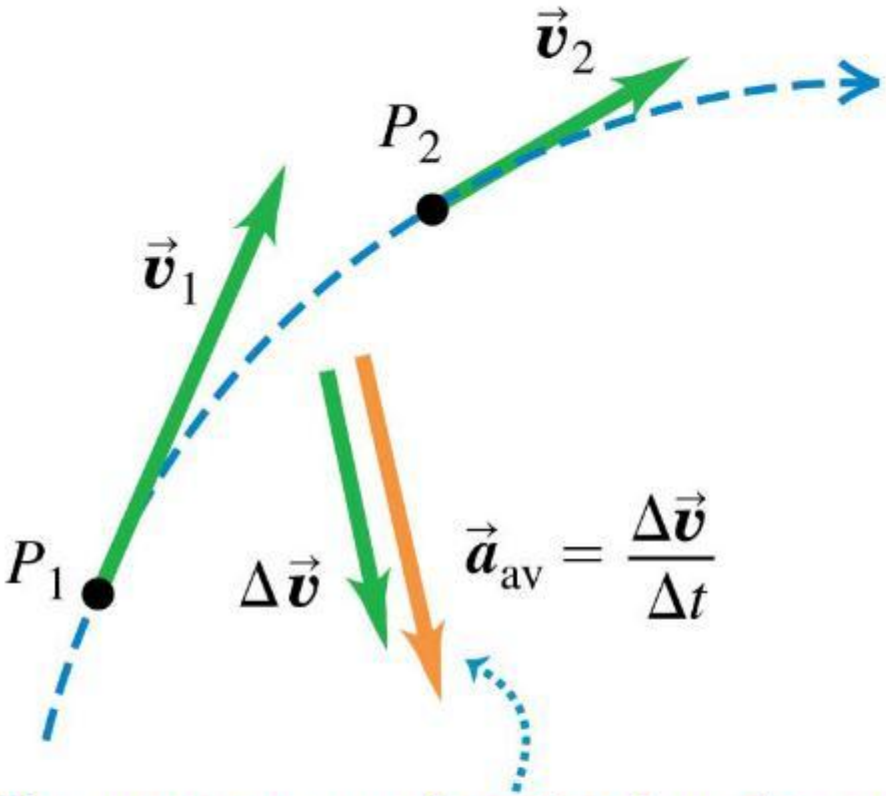
... and equals the instantaneous rate of change of its velocity vector.

Average acceleration

- The change in velocity between two points is determined by vector subtraction.



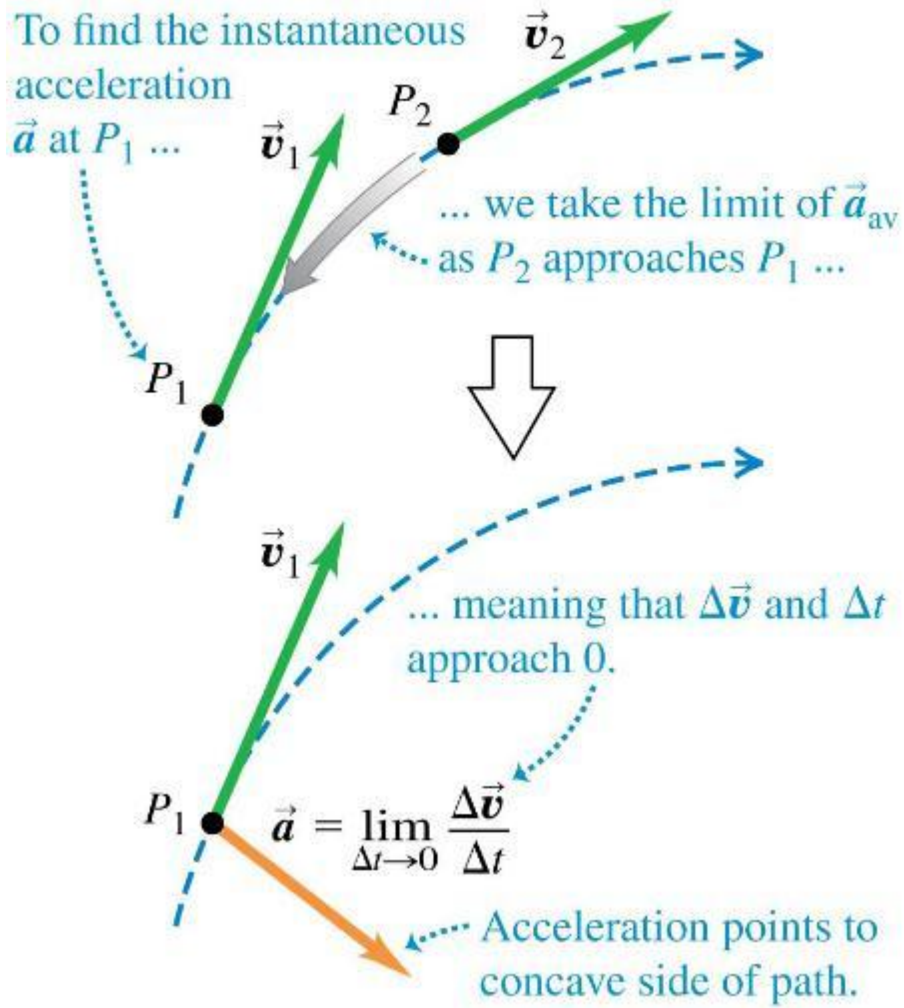
Average acceleration



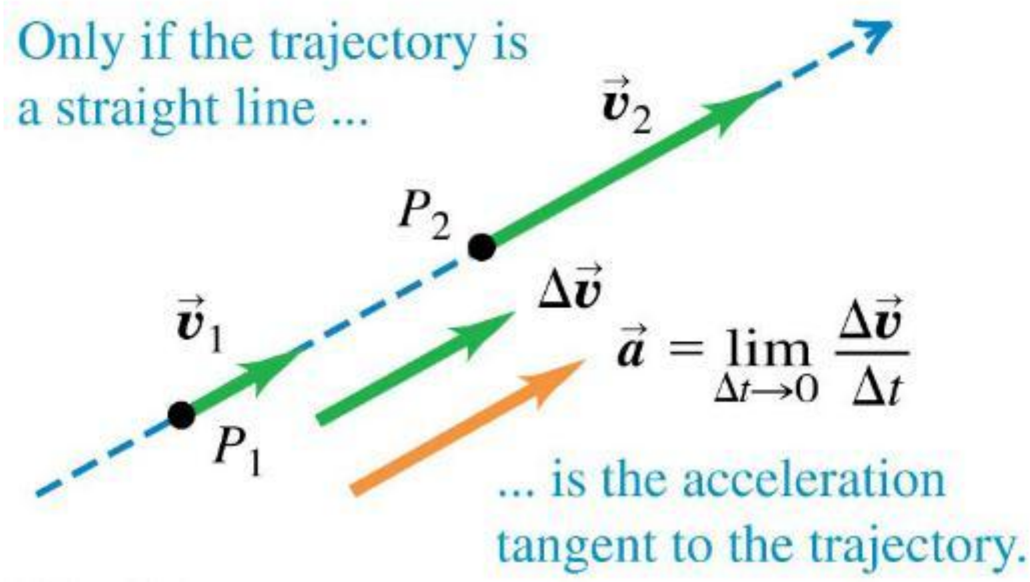
The average acceleration has the same direction as the change in velocity, $\Delta\vec{v}$.

Instantaneous acceleration

- The velocity vector is always tangent to the particle's path, but the instantaneous acceleration vector does *not* have to be tangent to the path.
- If the path is curved, the acceleration points toward the concave side of the path.



Instantaneous acceleration



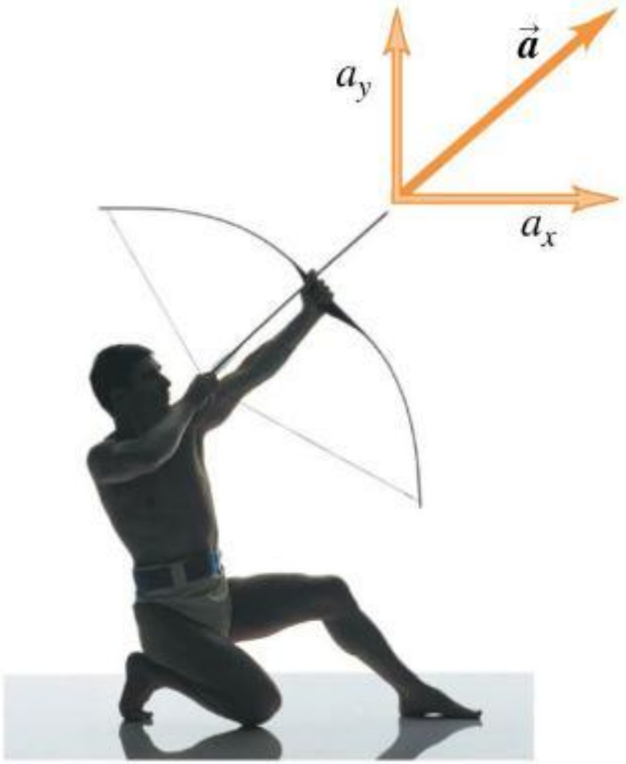
Components of acceleration

Each **component of** a particle's **instantaneous acceleration vector** ...

$$a_x = \frac{dv_x}{dt} \qquad a_y = \frac{dv_y}{dt} \qquad a_z = \frac{dv_z}{dt}$$

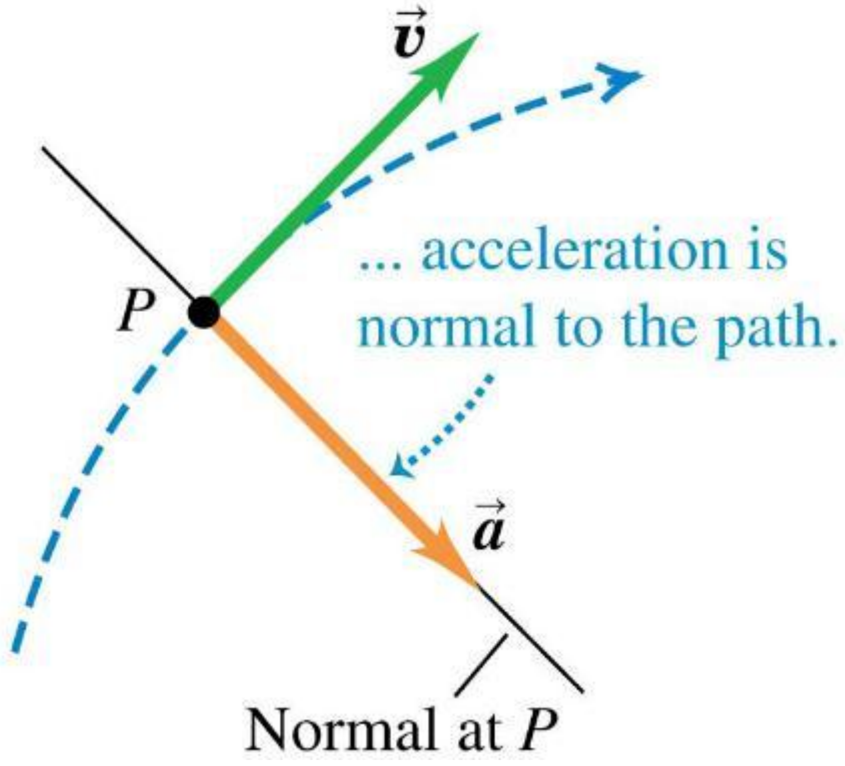
... equals the instantaneous rate of change of its corresponding velocity component.

- Shooting an arrow is an example of an acceleration vector that has both *x*- and *y*-components.



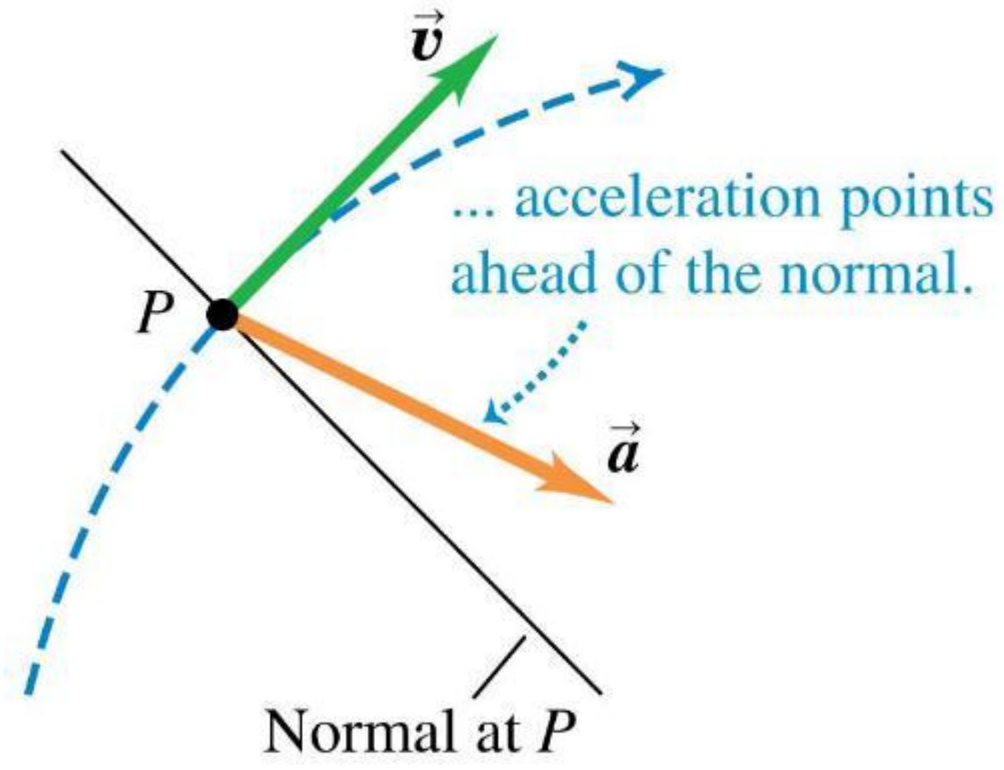
Parallel and perpendicular components of acceleration

- Velocity and acceleration vectors for a particle moving through a point P on a curved path with *constant speed*



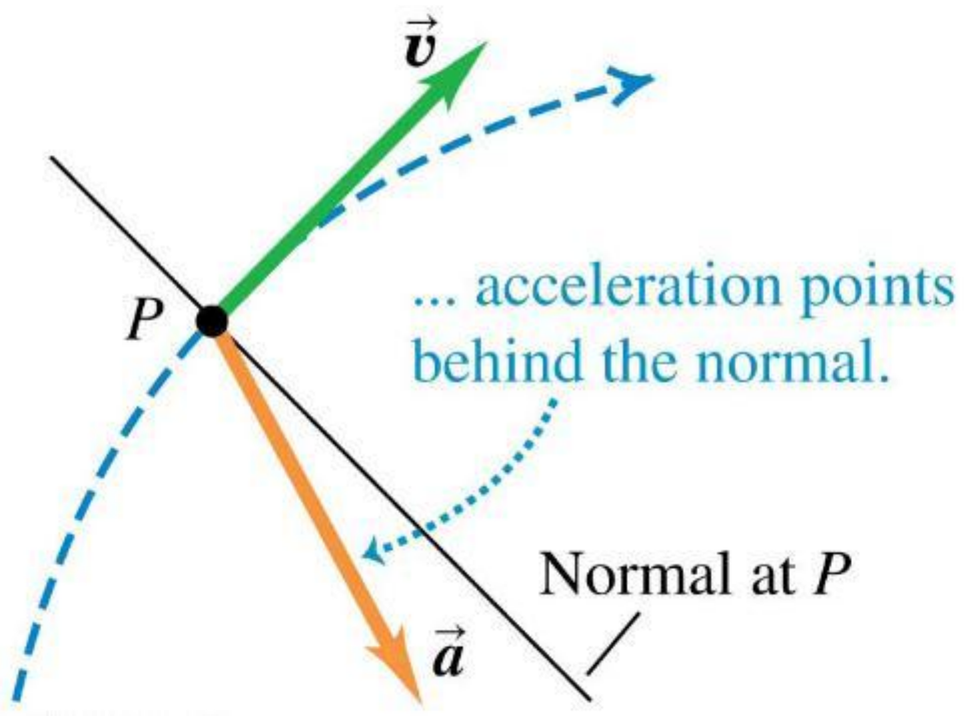
Parallel and perpendicular components of acceleration

- Velocity and acceleration vectors for a particle moving through a point *P* on a curved path with *increasing speed*



Parallel and perpendicular components of acceleration

- Velocity and acceleration vectors for a particle moving through a point P on a curved path with *decreasing speed*



Moon Man (2022)

The film centres around a maintenance man who is accidentally left alone on the moon after the operation to shatter the incoming asteroid, using the Moon to shield Earth from its fragments.

As a science fiction comedy, the possibly fictionalised, exaggerated, and unrealistic representations of certain basic laws of Physics become debatable and worth probing.

Questions

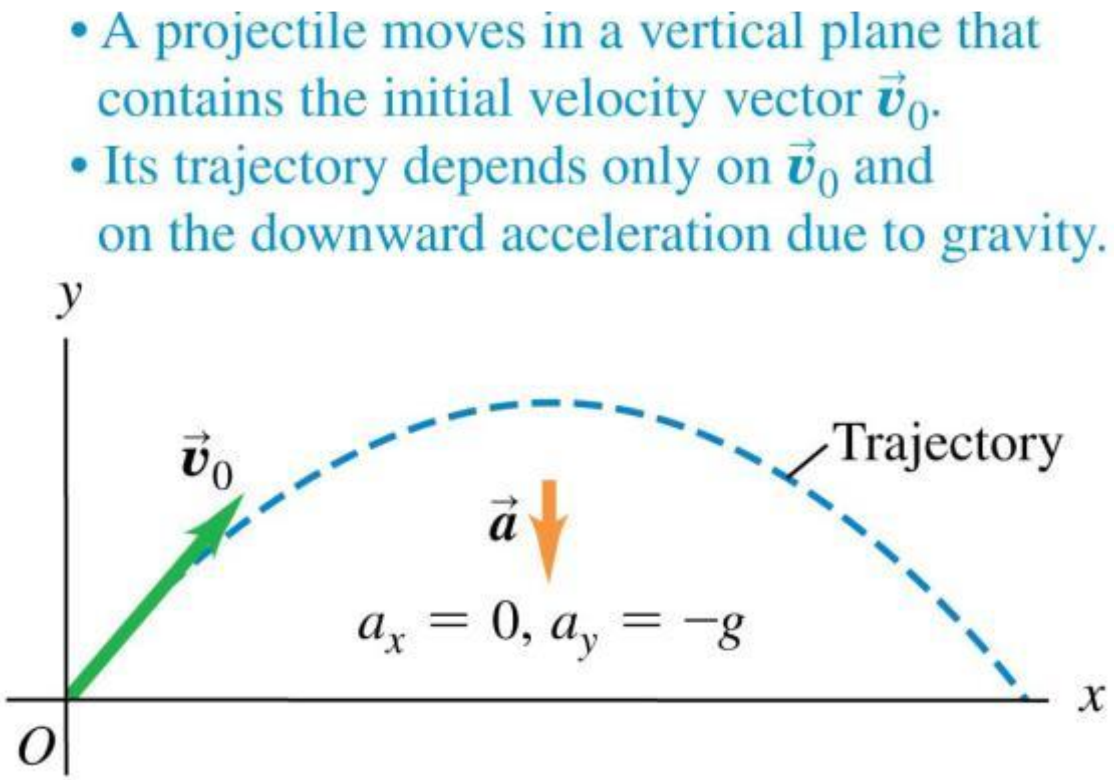
What are the differences between the Earth and Moon environments?

If we situate the scene on the Earth, do you think the spiky trajectory of the splashed dust we see still makes sense?

In your opinion, should the dust fly higher or lower on the moon (compared to on the Earth)?

Projectile motion

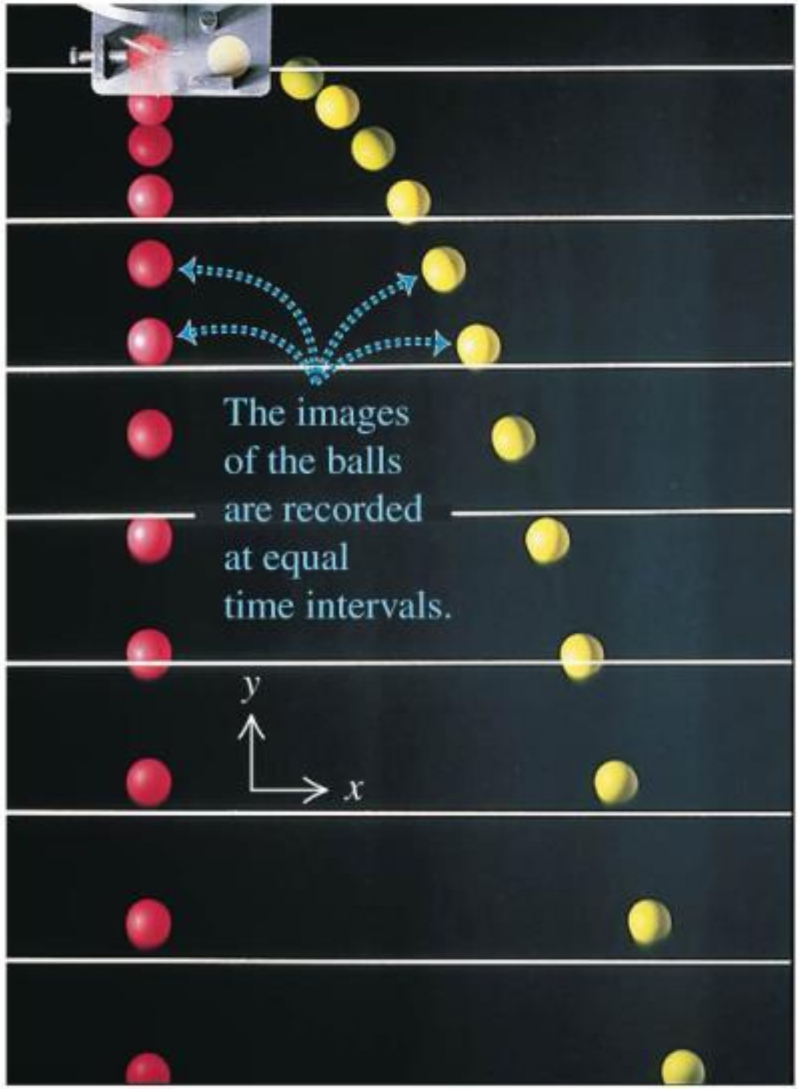
- A **projectile** is any body given an initial velocity that then follows a path determined by the effects of gravity and air resistance.
- Begin by neglecting resistance and the curvature and rotation of the earth.



The *x*- and *y*-motion are separable

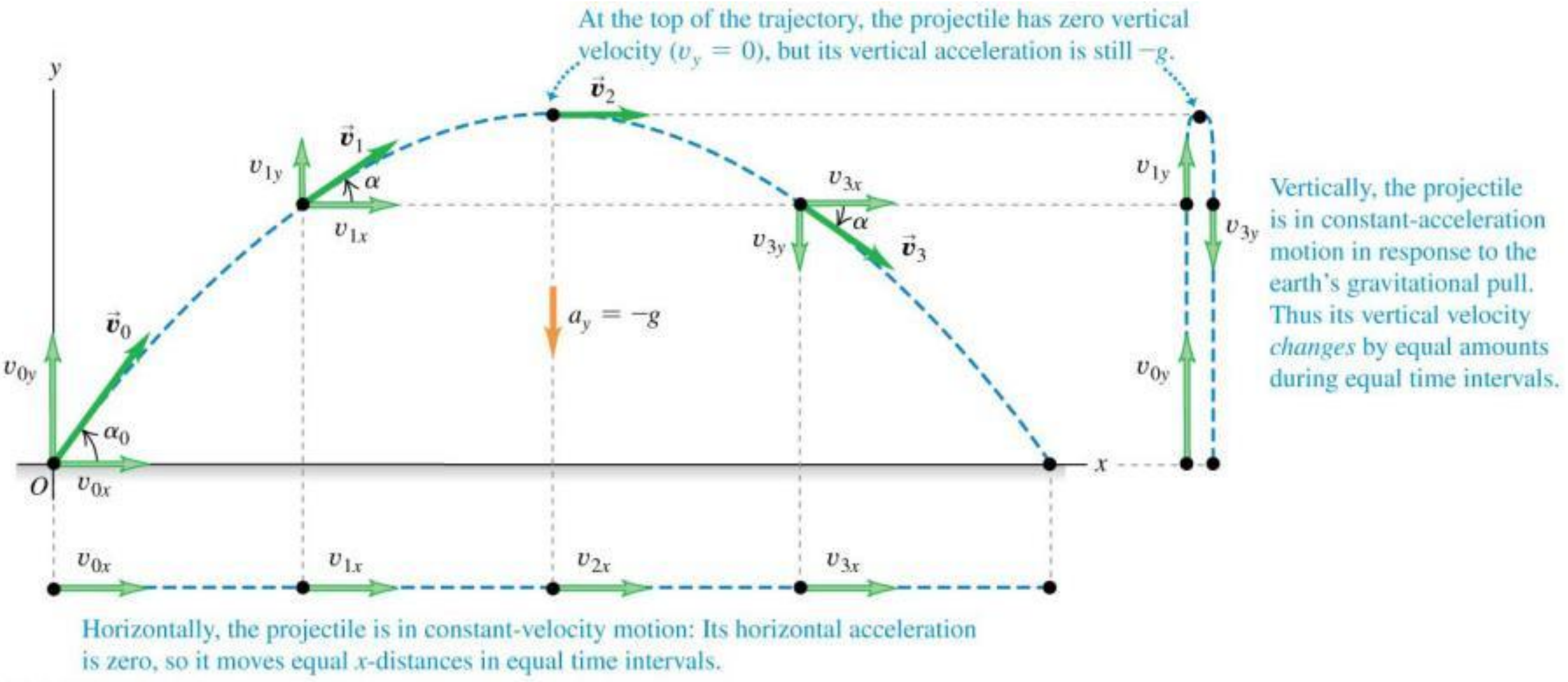
- The red ball is dropped at the same time that the yellow ball is fired horizontally.
- The strobe marks equal time intervals.
- We can analyze projectile motion as horizontal motion with constant velocity and vertical motion with constant acceleration:

$$a_x = 0 \text{ and } a_y = -g.$$



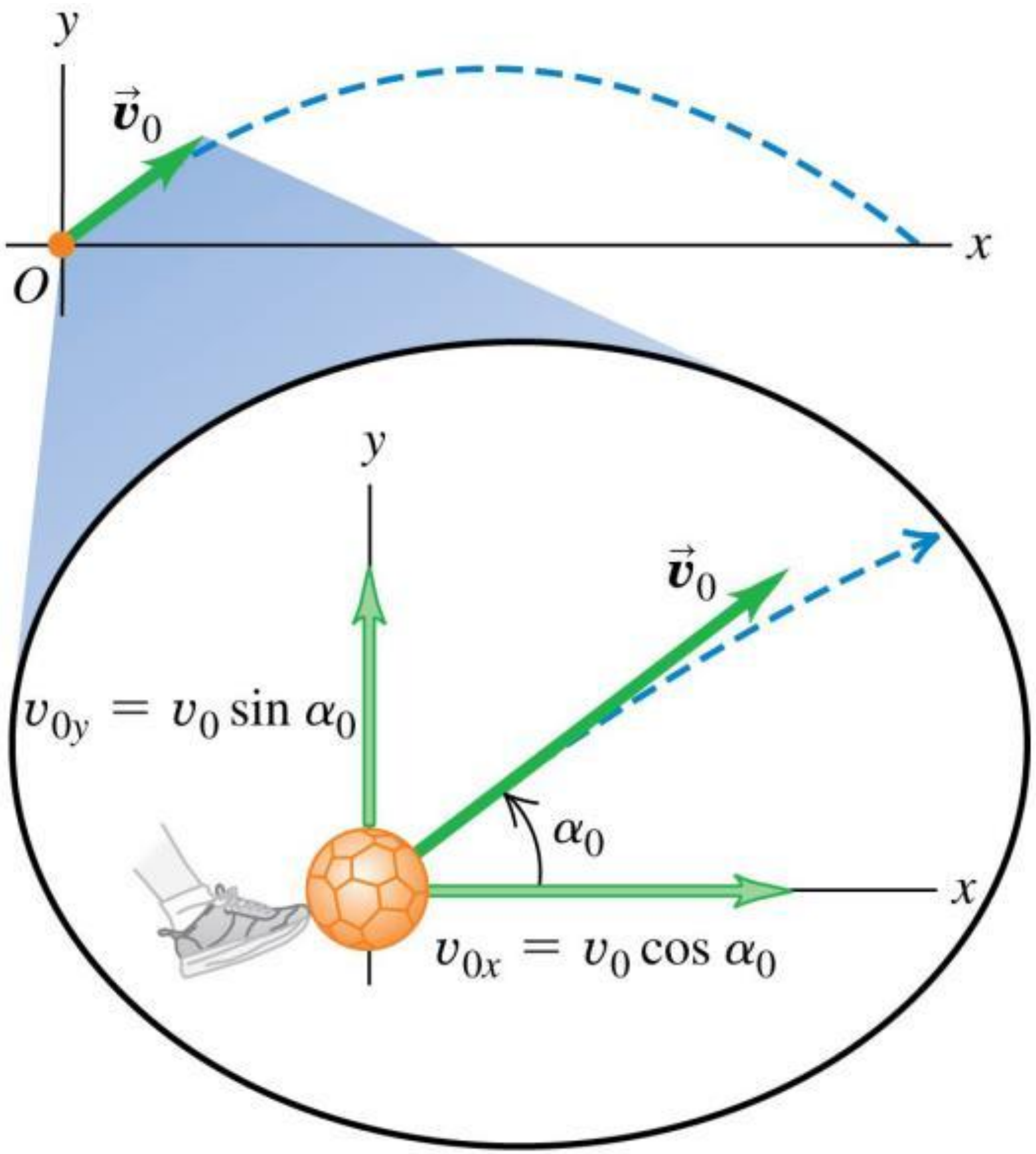
Projectile motion

- If air resistance is negligible, the trajectory of a projectile is a combination of horizontal motion with constant velocity and vertical motion with constant acceleration.

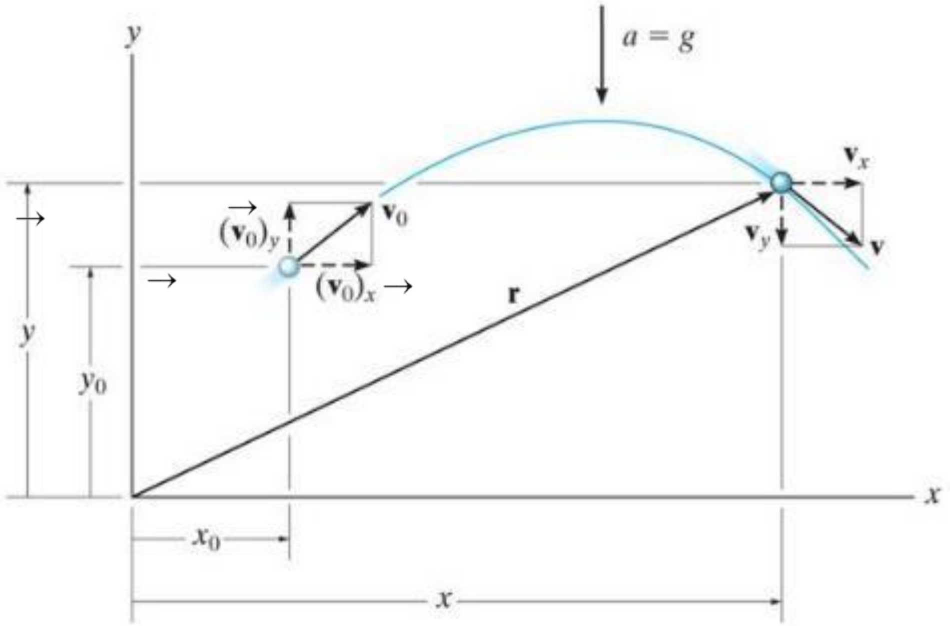


Projectile motion – Initial velocity

- The initial velocity components of a projectile (such as a kicked soccer ball) are related to the initial speed and initial angle.



Horizontal motion



Since $a_x = 0$, the velocity in the horizontal direction remains constant ($v_x = v_{ox}$) and the position in the x direction can be determined by:

$x = x_o + (v_{ox})\, t$

Why is a_x equal to zero (what assumption must be made if the movement is through the air)?

Vertical motion

Since the positive y-axis is directed upward, $a_y = -g$. Application of the constant acceleration equations yields:

$$v_y = v_{oy} - g\,t$$

$$y = y_o + (v_{oy})\,t - \tfrac{1}{2}\,g\,t^2$$

$$v_y^{\,2} = v_{oy}^{\,2} - 2\,g\,(y - y_o)$$

For any given problem, only two of these three equations can be used. Why?

The equations for projectile motion

- If we set $x_0 = y_0 = 0$, the equations describing projectile motion are shown below:

Coordinates at time t of a **projectile** (positive y-direction is upward, and $x = y = 0$ at $t = 0$)

$$x = (v_0 \cos \alpha_0)t$$

$$y = (v_0 \sin \alpha_0)t - \tfrac{1}{2}gt^2$$

Speed at $t = 0$

Direction at $t = 0$

Time

Acceleration due to gravity: Note $g > 0$.

Velocity components at time t of a **projectile** (positive y-direction is upward)

$$v_x = v_0 \cos \alpha_0$$

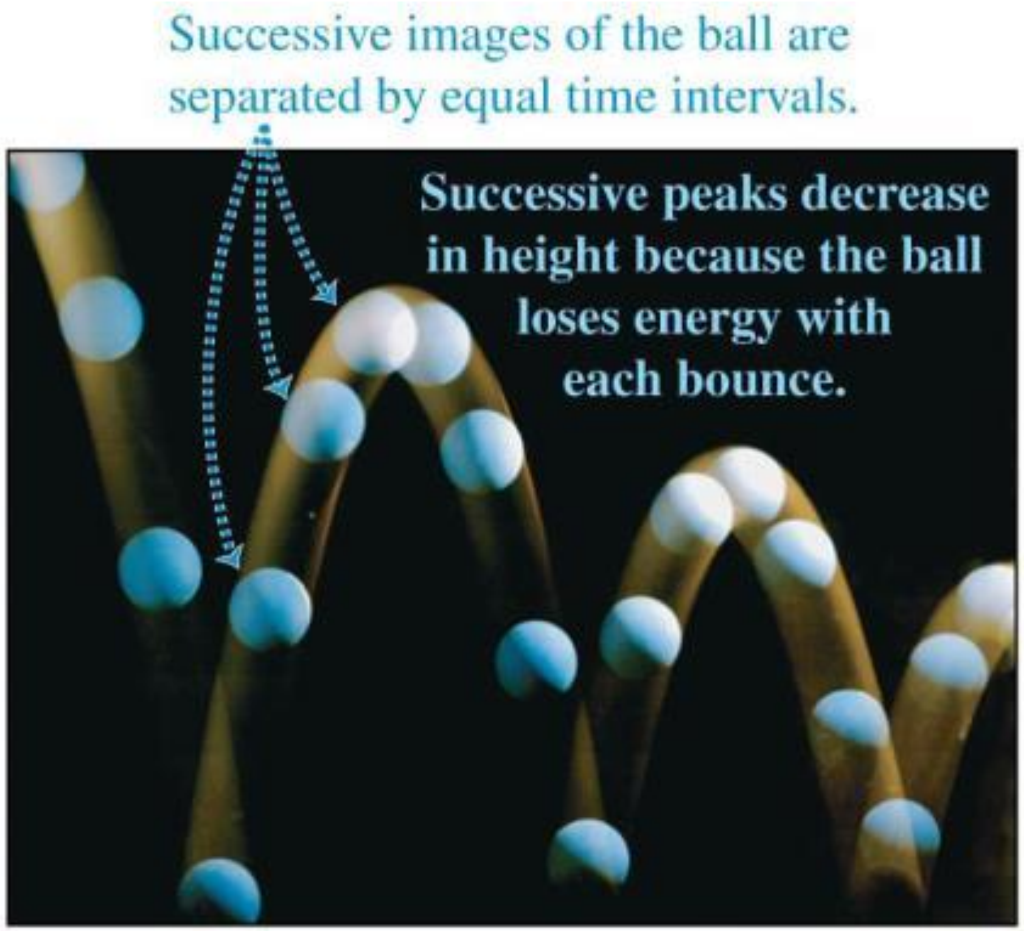
$$v_y = v_0 \sin \alpha_0 - gt$$

Speed at $t = 0$

Direction at $t = 0$

Time

Parabolic trajectories of a bouncing ball



Quiz

1. The downward acceleration of an object in free-flight motion is

   A) zero. B) increasing with time.

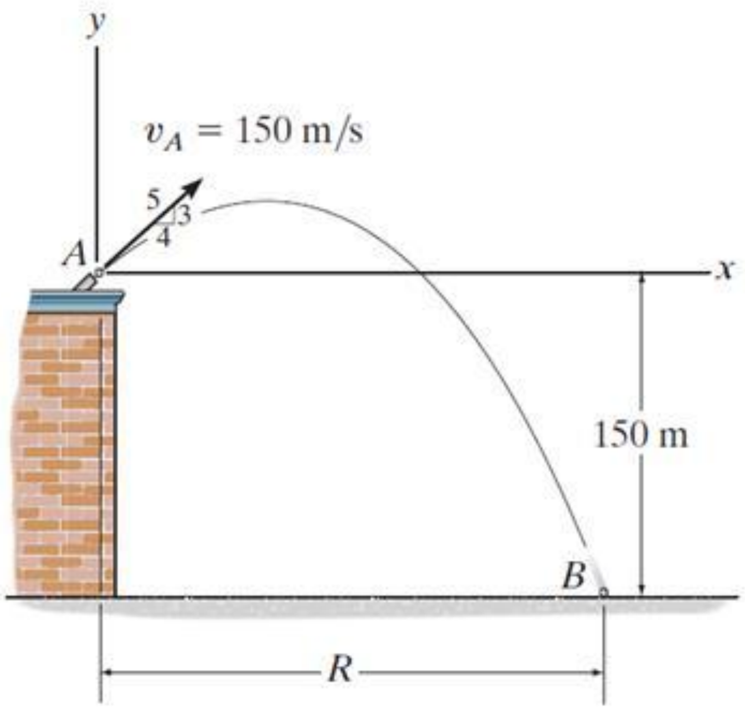
   C) 9.81 m/s^2. D) decreasing with time.

2. The horizontal component of velocity remains __________ during a free-flight motion.

   A) zero B) constant

   C) at 9.81 m/s^2 D) decreasing with time

Example



Given: Projectile is fired with v_A=150 m/s at point A.

Find: The horizontal distance it travels (R) and the time in the air.

Quiz

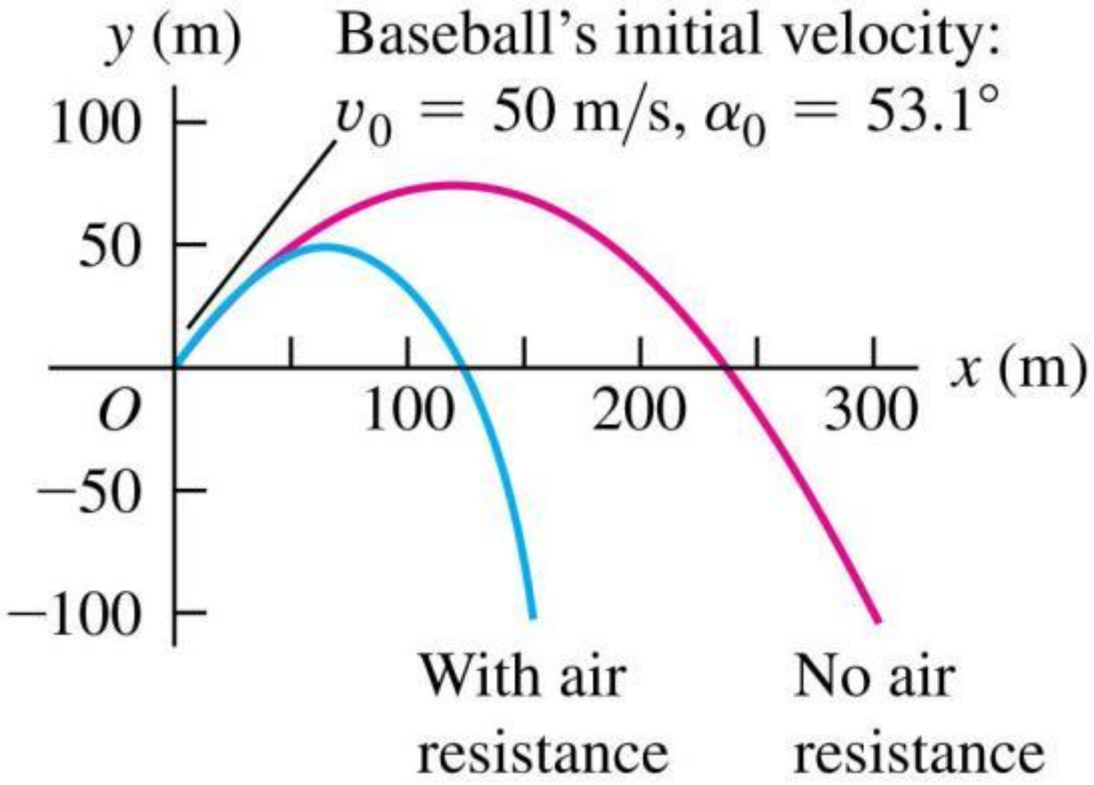
The time of flight of a projectile, fired over level ground, with initial velocity V_o at angle θ, is equal to?

A) $(v_o \sin \theta)/g$ B) $(2v_o \sin \theta)/g$

C) $(v_o \cos \theta)/g$ D) $(2v_o \cos \theta)/g$

The effects of air resistance

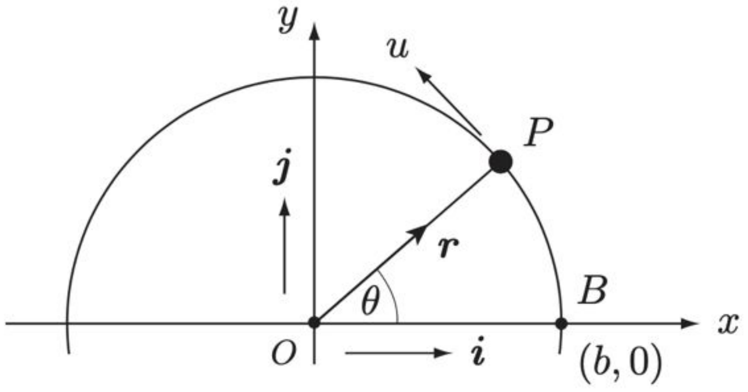
- Calculations become more complicated.
- Acceleration is not constant.
- Effects can be very large.
- Maximum height and range decrease.
- Trajectory is no longer a parabola.



Uniform circular motion

Consider a particle P moving with constant speed u in the anti-clockwise direction around a circle centre O and radius b. At time $t = 0$, P is at the point B(b, 0). What are its velocity and acceleration vectors at time t ?

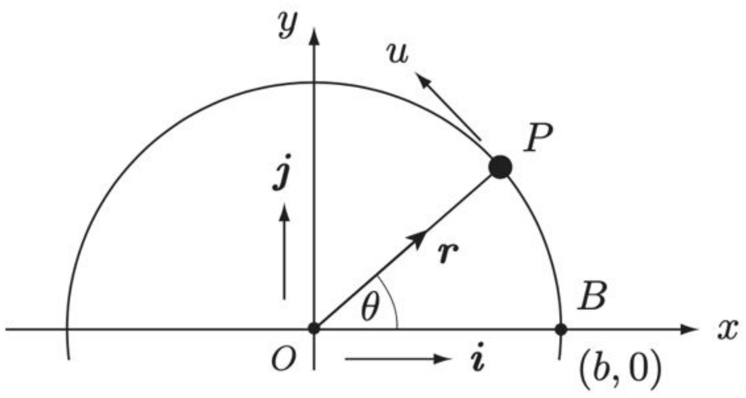
Since P moves with constant speed u, the arc length BP travelled in time t must be ut. The angle $\theta = ut/b$.



Uniform circular motion

The position vector of P at time t is therefore

$$\boldsymbol{r} = b\cos\theta\,\boldsymbol{i} + b\sin\theta\,\boldsymbol{j},$$
$$= b\cos(ut/b)\,\boldsymbol{i} + b\sin(ut/b)\,\boldsymbol{j}.$$



The velocity of P at time t is given by

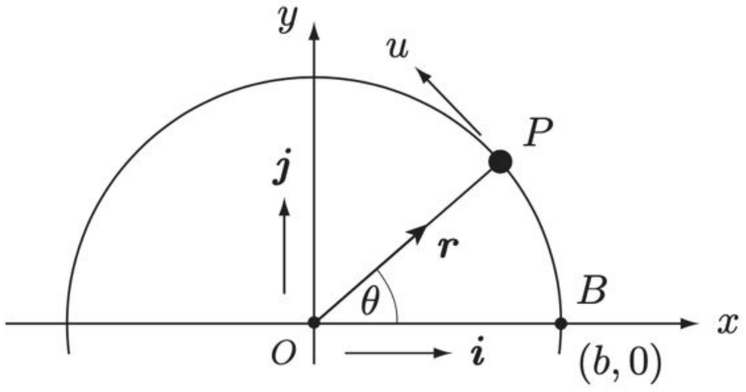
$$\boldsymbol{v} = \frac{d\boldsymbol{r}}{dt} = -u\sin(ut/b)\,\boldsymbol{i} + u\cos(ut/b)\,\boldsymbol{j},$$

$$|\boldsymbol{v}| = \left(u^2\cos^2(ut/b) + u^2\sin^2(ut/b)\right)^{1/2} = u$$

Uniform circular motion

The position vector of P at time t is therefore

$$\boldsymbol{r} = b\cos\theta\,\boldsymbol{i} + b\sin\theta\,\boldsymbol{j},$$
$$= b\cos(ut/b)\,\boldsymbol{i} + b\sin(ut/b)\,\boldsymbol{j}.$$



The acceleration of P at time t is given by

$$\boldsymbol{a} = \frac{d\boldsymbol{v}}{dt} = -\frac{u^2}{b}\cos(ut/b)\,\boldsymbol{i} - \frac{u^2}{b}\sin(ut/b)\,\boldsymbol{j}.$$

$$|\boldsymbol{a}| = \left(\left(\frac{u^2}{b}\right)^2\cos^2(ut/b) + \left(\frac{u^2}{b}\right)^2\sin^2(ut/b)\right)^{1/2} = \frac{u^2}{b}$$

$$\mathbf{a} = -\frac{u^2}{r}\mathbf{r}$$

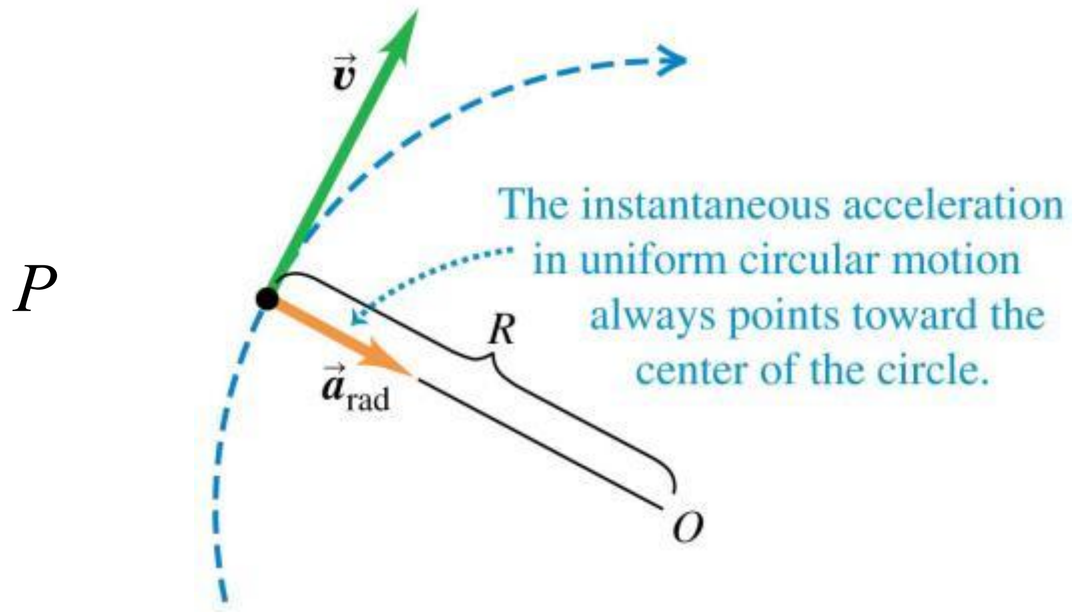
Acceleration for uniform circular motion

- For uniform circular motion, the instantaneous acceleration always points toward the center of the circle and is called the **centripetal acceleration**.

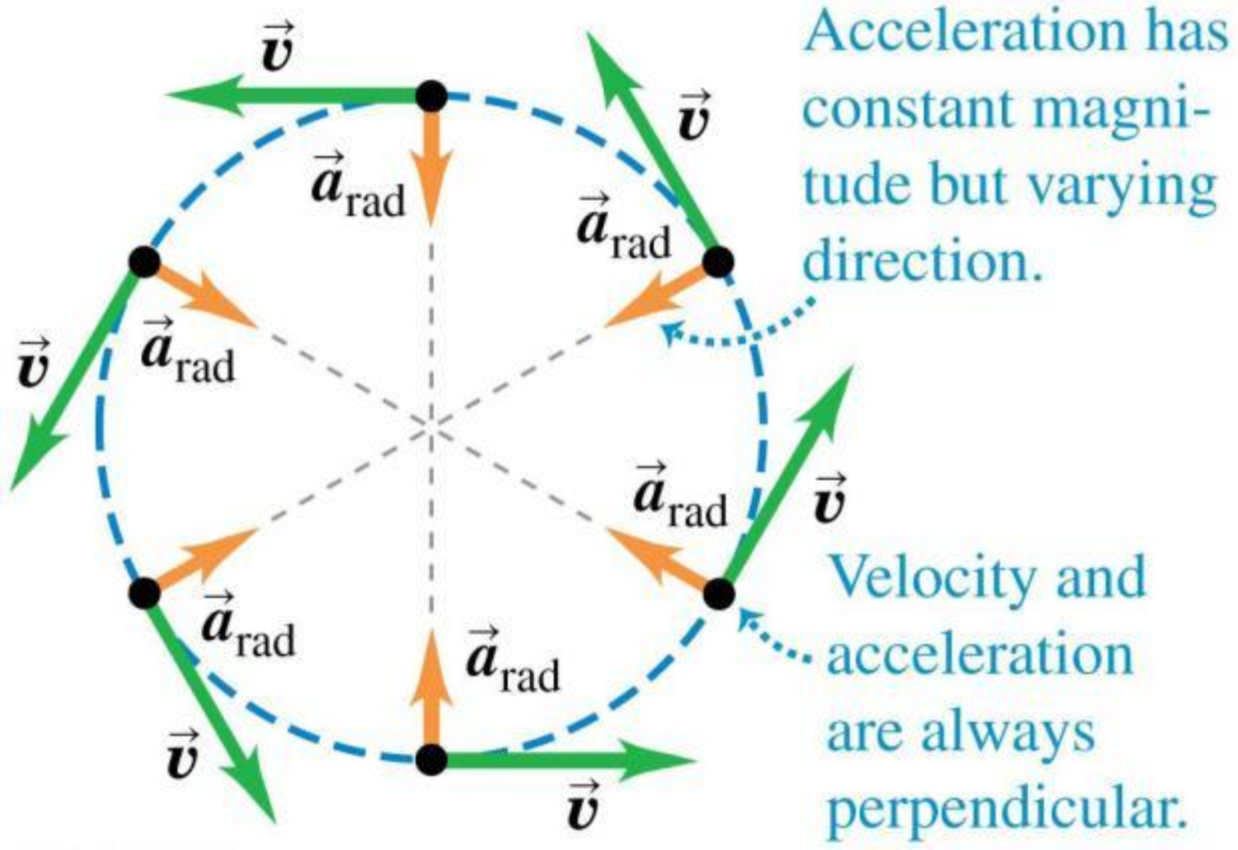
- The magnitude of the acceleration is $a_{\text{rad}} = v^2/R$.

- The direction is from P to O.

- The *period T* is the time for one revolution, and $a_{\text{rad}} = 4\pi^2 R/T^2$.

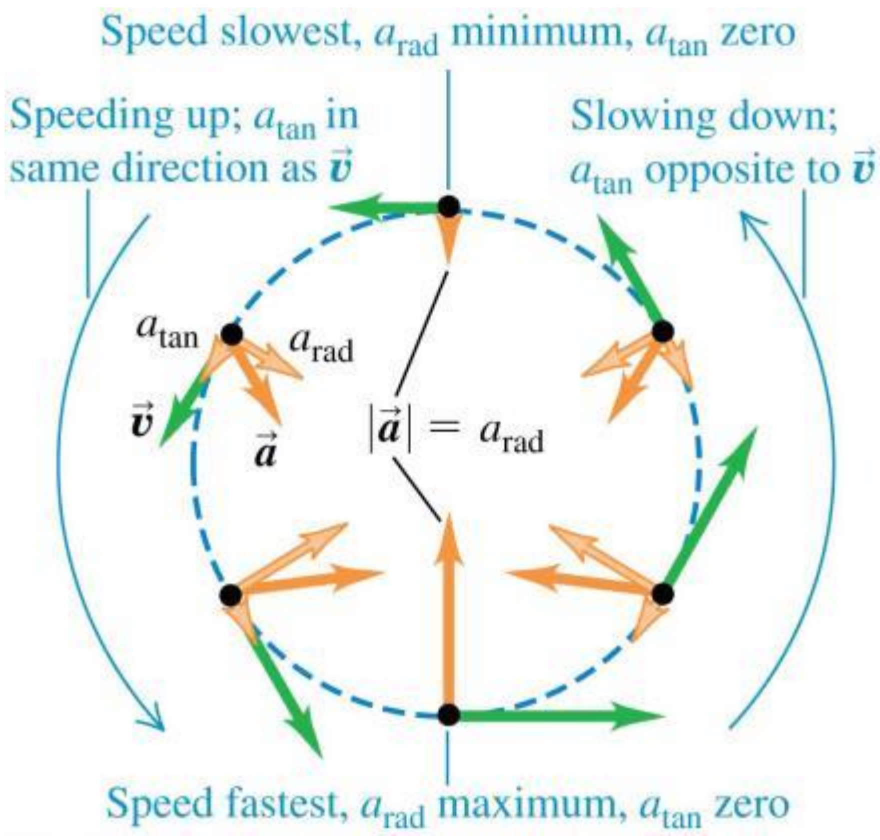


Uniform circular motion



Nonuniform circular motion

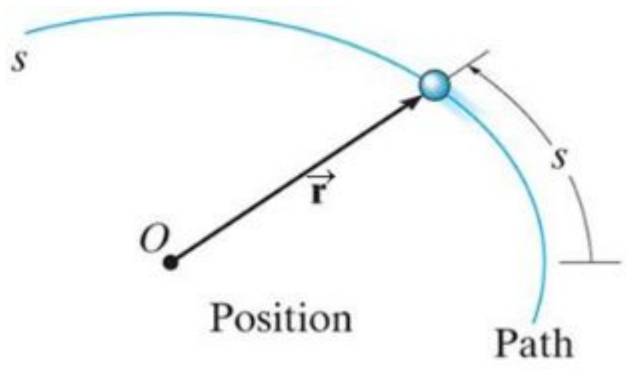
- If the speed varies, the motion is *nonuniform circular motion*.
- The radial acceleration component is still $a_{rad} = v^2/R$, but there is also a tangential acceleration component a_{tan} that is *parallel* to the instantaneous velocity.



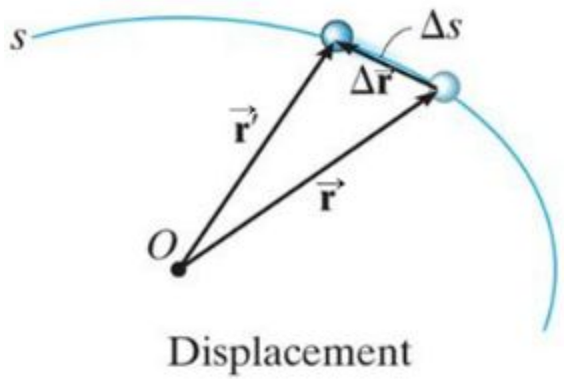
Curvilinear motion: general formalism

A particle moving along a curved path undergoes curvilinear motion. Since the motion is often three-dimensional, vectors are used to describe the motion.

A particle moves along a curve defined by the path function, s.



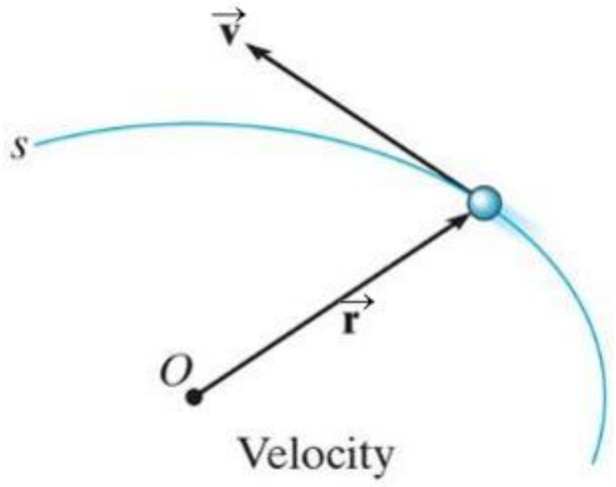
The position of the particle at any instant is designated by the vector $\vec{r} = r(t)$. Both the magnitude and direction of $\vec{r}$ may vary with time.



If the particle moves a distance Δs along the curve during time interval Δt, the displacement is determined by vector subtraction: $\Delta\vec{r} = \vec{r}\,' - \vec{r}$

Curvilinear motion: general formalism

Velocity represents the rate of change in the position of a particle.



The average velocity of the particle during the time increment Δt is $\vec{v}_{avg} = \Delta \vec{r} / \Delta t$.

The instantaneous velocity is the time-derivative of position $\vec{v} = d \vec{r} / dt$.

The velocity vector, $\boldsymbol{v}$, is always tangent to the path of motion.

The magnitude of $\vec{v}$ is called the speed. Since the arc length Δs approaches the magnitude of $\Delta \vec{r}$ as t$\rightarrow$0, the speed can be obtained by differentiating the path function (v = ds/dt). Note that this is not a vector!

Curvilinear motion: general formalism

Acceleration represents the rate of change in the velocity of a particle.

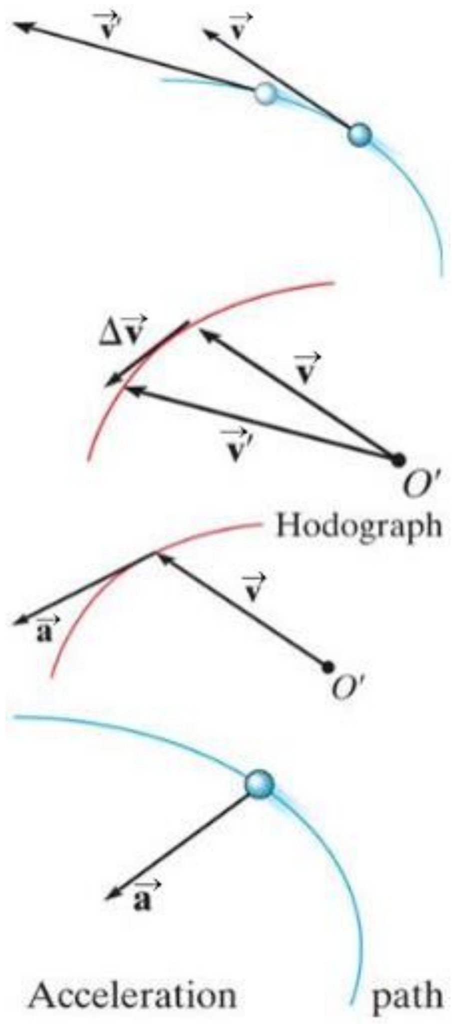
If a particle's velocity changes from $\boldsymbol{v}$ to $\boldsymbol{v}'$ over a time increment Δt, the average acceleration during that increment is:

$$\vec{a}_{avg} = \Delta \vec{v} / \Delta t = (\vec{v}' - \vec{v}) / \Delta t$$

The instantaneous acceleration is the time-derivative of velocity:

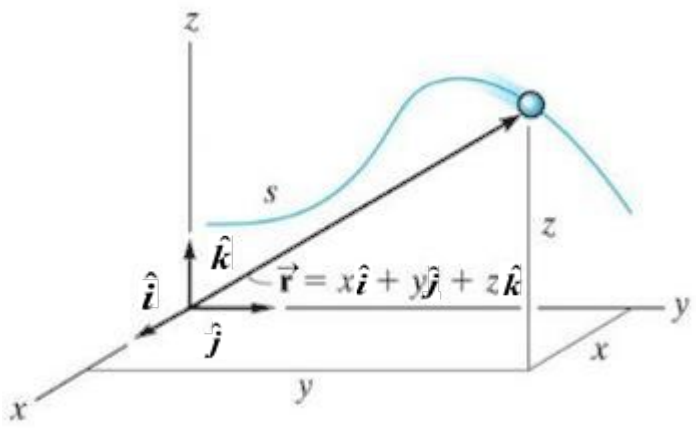
$$\vec{a} = d \vec{v} / dt = d^2 \vec{r} / dt^2$$

A plot of the locus of points defined by the arrowhead of the velocity vector is called a hodograph. The acceleration vector is tangent to the hodograph, but not, in general, tangent to the path function.



Curvilinear motion: rectangular components

It is often convenient to describe the motion of a particle in terms of its x, y, z or rectangular components, relative to a fixed frame of reference. This is the so-called "Cartesian coordinate".



Position

The position of the particle can be defined at any instant by the position vector

$$\vec{r} = x\,\hat{i} + y\,\hat{j} + z\,\hat{k}\ .$$

The x, y, z components may all be functions of time, i.e., $x = x(t)$, $y = y(t)$, and $z = z(t)$.

The magnitude of the position vector is: $r = (x^2 + y^2 + z^2)^{1/2}$

The direction of $\boldsymbol{r}$ is defined by the unit vector: $\vec{u}_r = (1/r)\vec{r}$

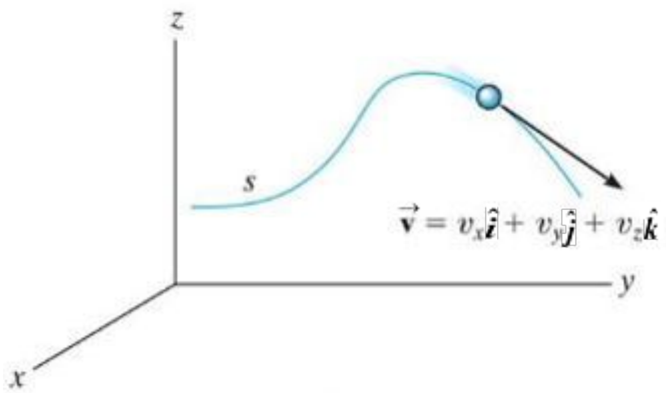
Rectangular components: velocity

The velocity vector is the time derivative of the position vector:

$$\vec{v} = d\,\vec{r}\,/dt = d(x\,\hat{i}\,)/dt + d(y\hat{j}\,)/dt + d(z\hat{k}\,)/dt$$

Since the unit vectors $\hat{i}$, $\hat{j}$, $\hat{k}$ are constant in magnitude and direction, this equation reduces to $\vec{v} = v_x\hat{i} + v_y\hat{j} + v_z\hat{k}$

where $v_x = \dot{x} = dx/dt$, $v_y = \dot{y} = dy/dt$, $v_z = \dot{z} = dz/dt$



Velocity

The magnitude of the velocity vector is

$$v = [(v_x)^2 + (v_y)^2 + (v_z)^2]^{1/2}$$

The direction of $\vec{v}$ is tangent to the path of motion.

Rectangular components: acceleration

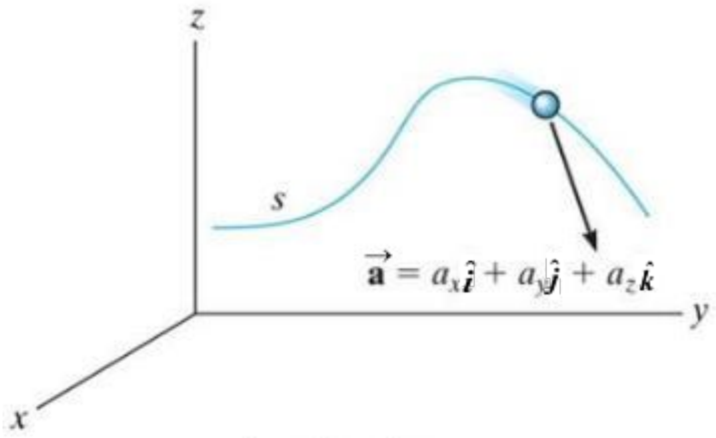
The acceleration vector is the time derivative of the velocity vector (second derivative of the position vector):

$$\vec{a} = d\,\vec{v}\,/dt = d^2\,\vec{r}\,/dt^2 = a_x\,\hat{\imath} + a_y\,\hat{\jmath} + a_z\,\hat{k}$$

where

$$a_x = \ddot{x} = \dot{v}_x = \frac{dv_x}{dt} \qquad a_y = \ddot{y} = \dot{v}_y = \frac{dv_y}{dt} \qquad a_z = \ddot{z} = \dot{v}_z = \frac{dv_z}{dt}$$

The magnitude of the acceleration vector is $a = [(a_x)^2 + (a_y)^2 + (a_z)^2\,]^{0.5}$



Acceleration

The direction of $\vec{a}$ is usually not tangent to the path of the particle.

Quiz

1. In curvilinear motion, the direction of the instantaneous velocity is always

   A) tangent to the hodograph.
   B) perpendicular to the hodograph.
   C) tangent to the path.
   D) perpendicular to the path.

2. In curvilinear motion, the direction of the instantaneous acceleration is always

   A) tangent to the hodograph.
   B) perpendicular to the hodograph.
   C) tangent to the path.
   D) perpendicular to the path.

Example

Given: The box slides down the slope described by the equation $y = (0.05x^2)$ m, where x is in meters.
$v_x = -3$ m/s, $a_x = -1.5$ m/s^2 at $x = 5$ m.

Find: The y components of the velocity and the acceleration of the box at $x = 5$ m.

Quiz

1. If the position of a particle is defined by
   $\vec{r} = [(1.5t^2 + 1)\,\hat{\imath} + (4t - 1)\,\hat{\jmath}\,]$ (m), its speed at $t = 1$ s is

   A) 2 m/s B) 3 m/s

   C) 5 m/s D) 7 m/s

2. The path of a particle is defined by $y = 0.5x^2$. If the component of its velocity along the x-axis at $x = 2$ m is
   $v_x = 1$ m/s, its velocity component along the y-axis at this position is ____.

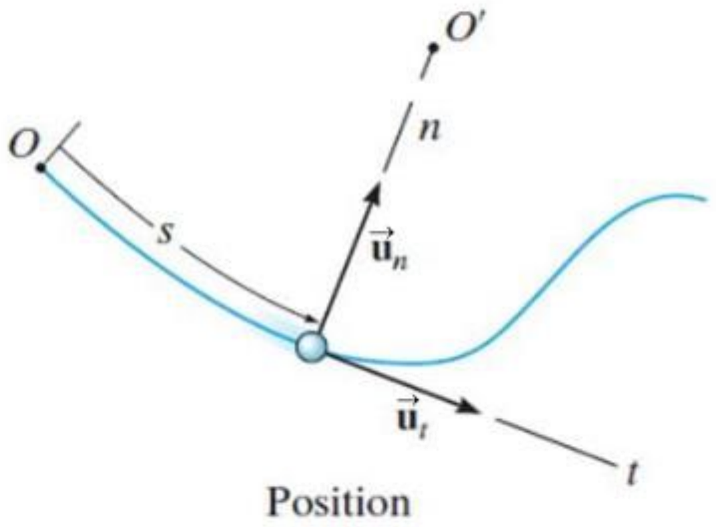
   A) 0.25 m/s B) 0.5 m/s

   C) 1 m/s D) 2 m/s

Curvilinear motion: n-t coordinate system

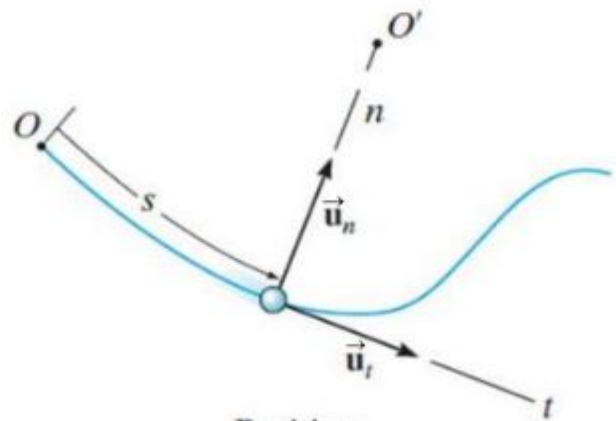
When a particle moves along a curved path, it is sometimes convenient to describe its motion using coordinates other than Cartesian. When the path of motion is known, normal (n) and tangential (t) coordinates are often used.

In the n-t coordinate system, the origin is located on the particle (the origin moves with the particle).

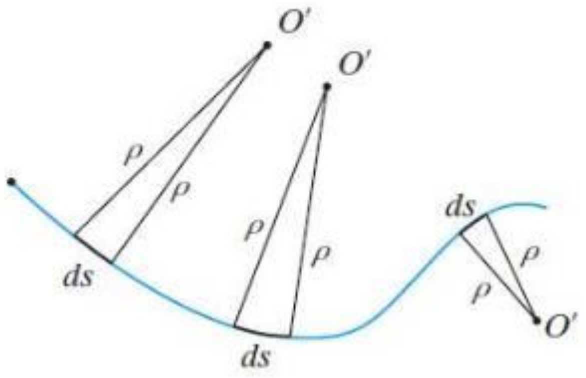


The t-axis is tangent to the path (curve) at the instant considered, positive in the direction of the particle's motion.
The n-axis is perpendicular to the t-axis with the positive direction toward the center of curvature of the curve.

Curvilinear motion: n-t coordinate system



Position



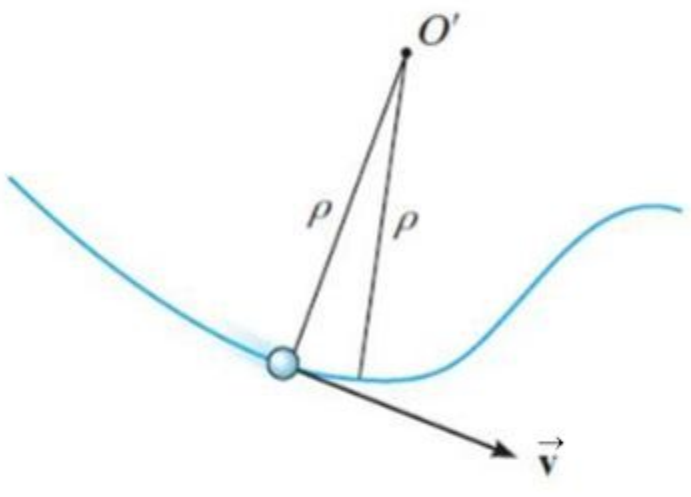
Radius of curvature

The positive n and t directions are defined by the unit vectors $\vec{u}_n$ and $\vec{u}_t$, respectively.

The center of curvature, O', always lies on the concave side of the curve.
The radius of curvature, r, is defined as the perpendicular distance from the curve to the center of curvature at that point.

The position of the particle at any instant is defined by the distance, s, along the curve from a fixed reference point.

Velocity in the n-t coordinate system



Velocity

The velocity vector is <u>always</u> tangent to the path of motion (t-direction).

The magnitude is determined by taking the time derivative of the path function, s(t).

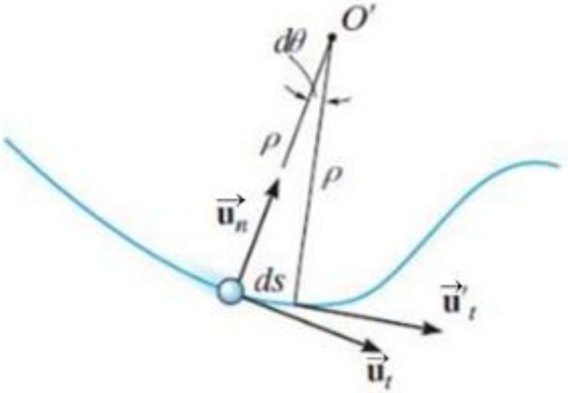
$$\vec{v} = v\,\vec{u}_t \qquad \text{where} \qquad v = \dot{s} = ds/dt$$

Here v defines the magnitude of the velocity (speed) and $\vec{u}_t$ defines the direction of the velocity vector.

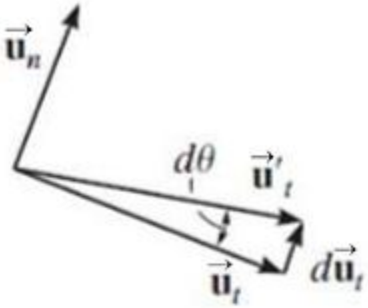
Acceleration in the n-t coordinate system

Acceleration is the time rate of change of velocity:

$$\vec{a} = d\,\vec{v}/dt = d(v\,\vec{u}_t)/dt = \dot{v}\vec{u}_t + v\,\dot{\vec{u}}_t$$



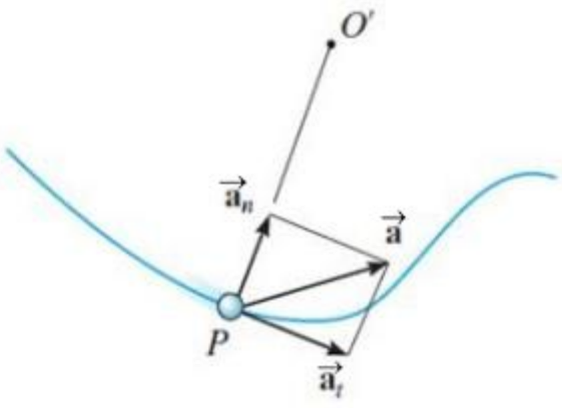
Here $\dot{v}$ represents the change in the magnitude of velocity and $\dot{\vec{u}}_t$ represents the rate of change in the direction of $\vec{u}_t$.



After mathematical manipulation, the acceleration vector can be expressed as:

$$\vec{a} = \dot{v}\,\vec{u}_t + (v^2/\rho)\,\vec{u}_n = a_t\,\vec{u}_t + a_n\,\vec{u}_n.$$

Acceleration in the n-t coordinate system



Acceleration

So, there are two components to the acceleration vector:

$$\vec{a} = a_t \vec{u}_t + a_n \vec{u}_n$$

- The tangential component is tangent to the curve and in the direction of increasing or decreasing velocity.

  $a_t = \dot{v}$ or $a_t\, ds = v\, dv$

- The normal or centripetal component is always directed toward the center of curvature of the curve. $a_n = v^2/\rho$

- The magnitude of the acceleration vector is

  $a = [(a_t)^2 + (a_n)^2]^{1/2}$

Special cases of motion

There are some special cases of motion to consider.

1) The particle moves along a straight line.

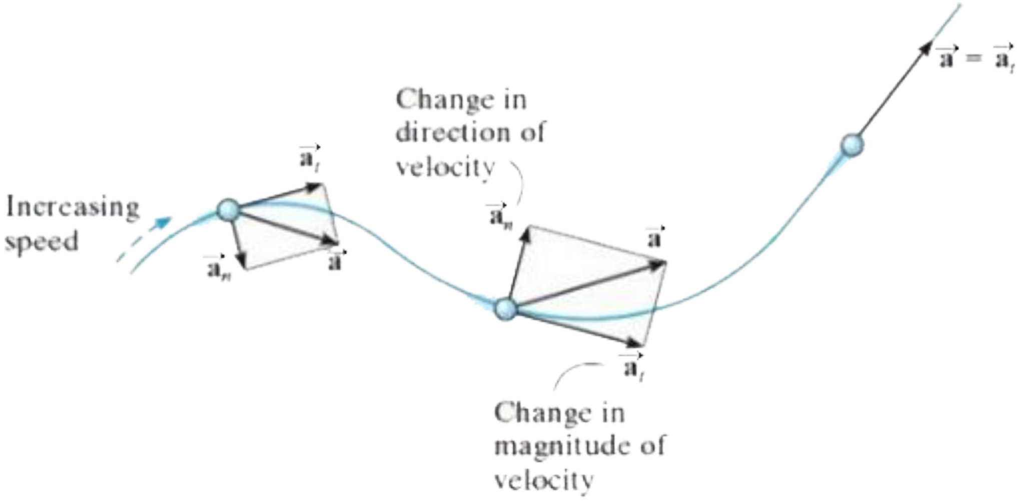
$$r = \infty \quad \Rightarrow \quad a_n = v^2/\rho = 0 \quad \Rightarrow \quad a = a_t = \dot{v}$$

The tangential component represents the time rate of change in the magnitude of the velocity.

2) The particle moves along a curve at constant speed.

$$a_t = \dot{v} = 0 \quad \Rightarrow \quad a = a_n = v^2/\rho$$

The normal component represents the time rate of change in the direction of the velocity.



Special cases of motion

3) The tangential component of acceleration is constant, $a_t = (a_t)_c$.

In this case,

$s = s_o + v_o t + (1/2)(a_t)_c t^2$

$v = v_o + (a_t)_c t$

$v^2 = (v_o)^2 + 2(a_t)_c (s - s_o)$

As before, s_o and v_o are the initial position and velocity of the particle at $t = 0$. How are these equations related to projectile motion equations? Why?

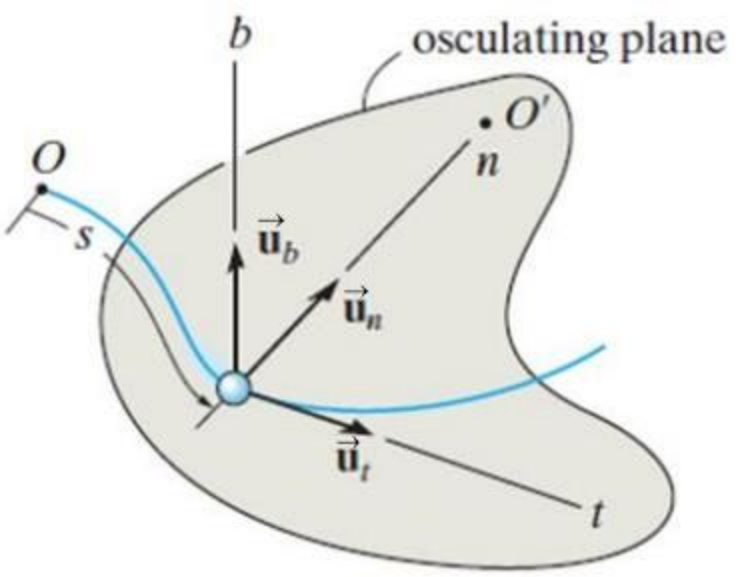
4) The particle moves along a path expressed as $y = f(x)$.

The radius of curvature, ρ, at any point on the path can be calculated from

$$\rho = \frac{\left[1 + (dy/dx)^2\right]^{\frac{3}{2}}}{|d^2y/dx^2|}$$

Derivation of the radius of curvature

Three-dimensional motion



If a particle moves along a space curve, the n and t axes are defined as before. At any point, the t-axis is tangent to the path and the n-axis points toward the center of curvature. The plane containing the n and t axes is called the osculating plane.

A third axis can be defined, called the binomial axis, b. The binomial unit vector, $\vec{u}_b$, is directed perpendicular to the osculating plane, and its sense is defined by the cross product $\vec{u}_b = \vec{u}_t \times \vec{u}_n$.

There is no motion, thus no velocity or acceleration, in the binomial direction.

Quiz

1. If a particle moves along a curve with a constant speed, then its tangential component of acceleration is

   A) positive. B) negative.

   C) zero. D) constant.

2. The normal component of acceleration represents

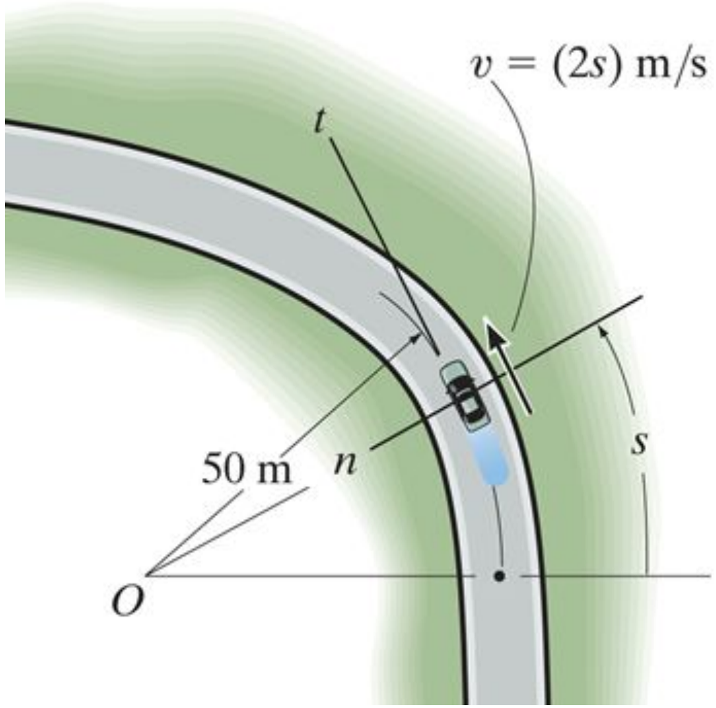
   A) the time rate of change in the magnitude of the velocity.

   B) the time rate of change in the direction of the velocity.

   C) magnitude of the velocity.

   D) direction of the total acceleration.

Example 1



Given: A car travels along the road with a speed of v = (2s) m/s, where s is in meters.

$\rho = 50$ m

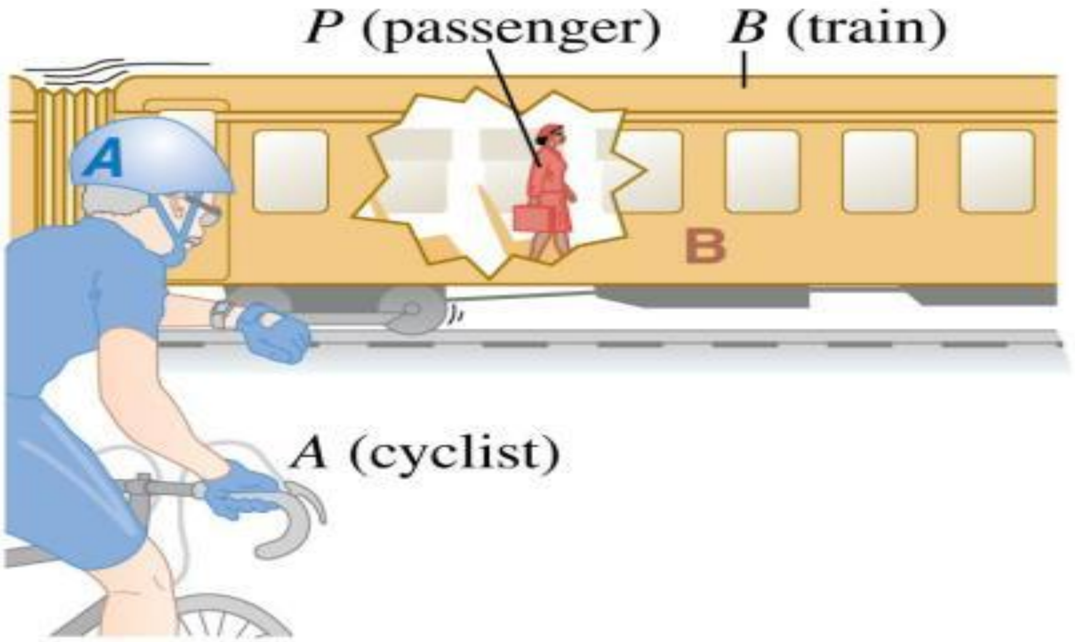
Find: The magnitudes of the car's acceleration at s = 10 m.

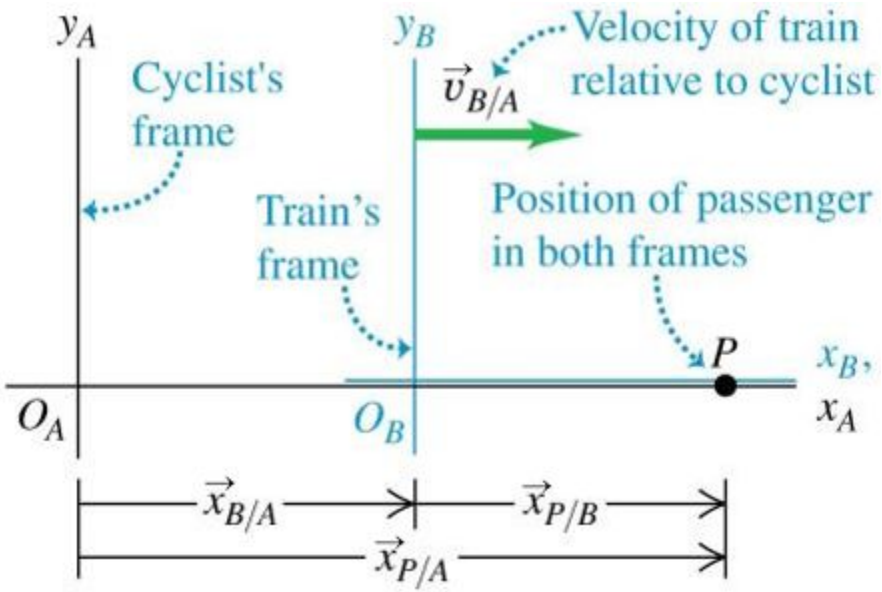
Relative velocity

- The velocity of a moving body seen by a particular observer is called the velocity *relative* to that observer, or simply the **relative velocity**.
- A **frame of reference** is a coordinate system plus a time scale.
- In many situations relative velocity is extremely important.

Relative velocity in one dimension

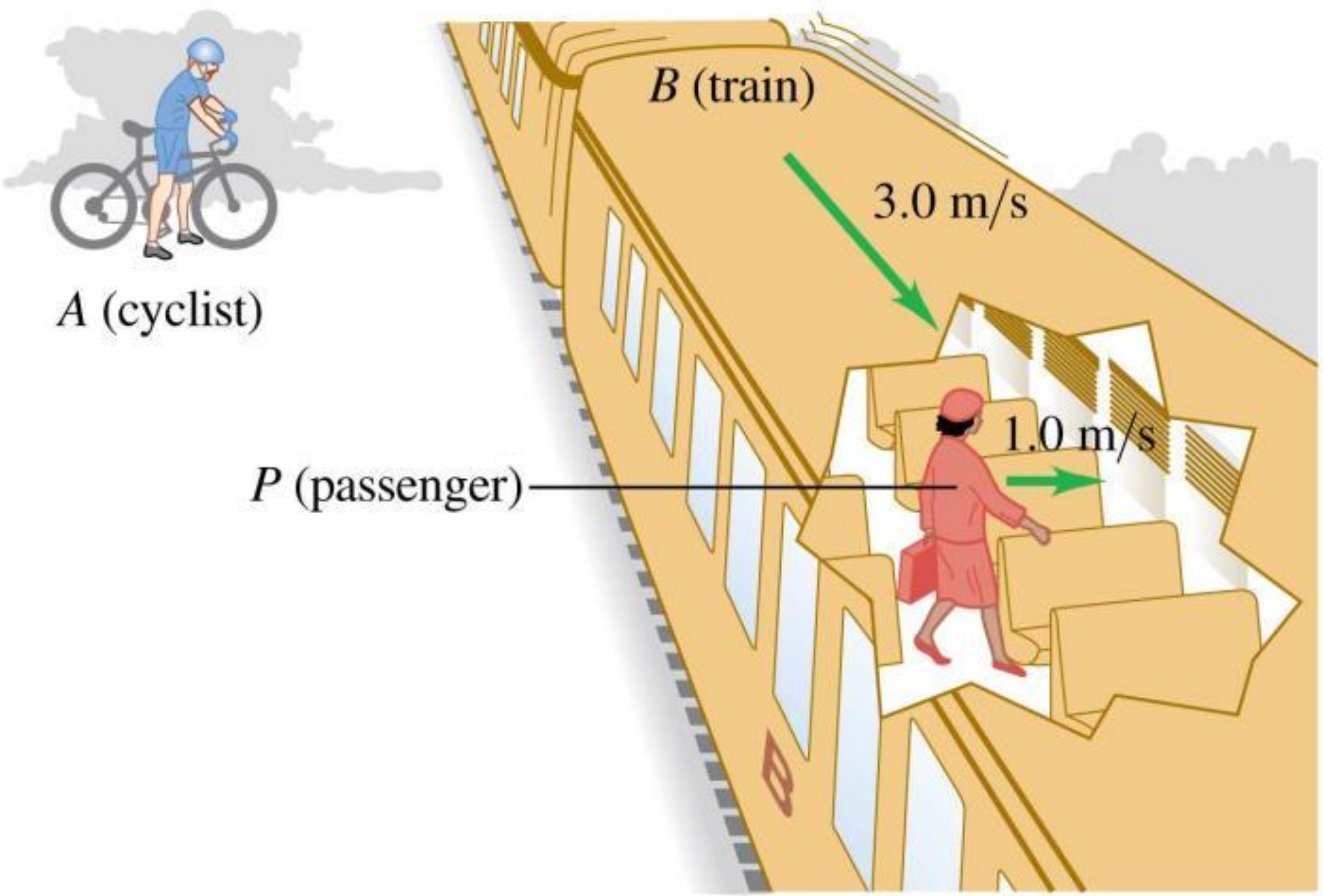
- If point P is moving relative to reference frame A, we denote the velocity of P relative to frame A as $\vec{\boldsymbol{v}}_{P/A}$.
- If P is moving relative to frame B and frame B is moving relative to frame A, then $\vec{\boldsymbol{x}}_{P/A\text{-}x} = \vec{\boldsymbol{x}}_{P/B\text{-}x} + \vec{\boldsymbol{x}}_{B/A\text{-}x}$. The x-velocity of P relative to frame A is $\vec{\boldsymbol{v}}_{P/A\text{-}x} = \vec{\boldsymbol{v}}_{P/B\text{-}x} + \vec{\boldsymbol{v}}_{B/A\text{-}x}$.



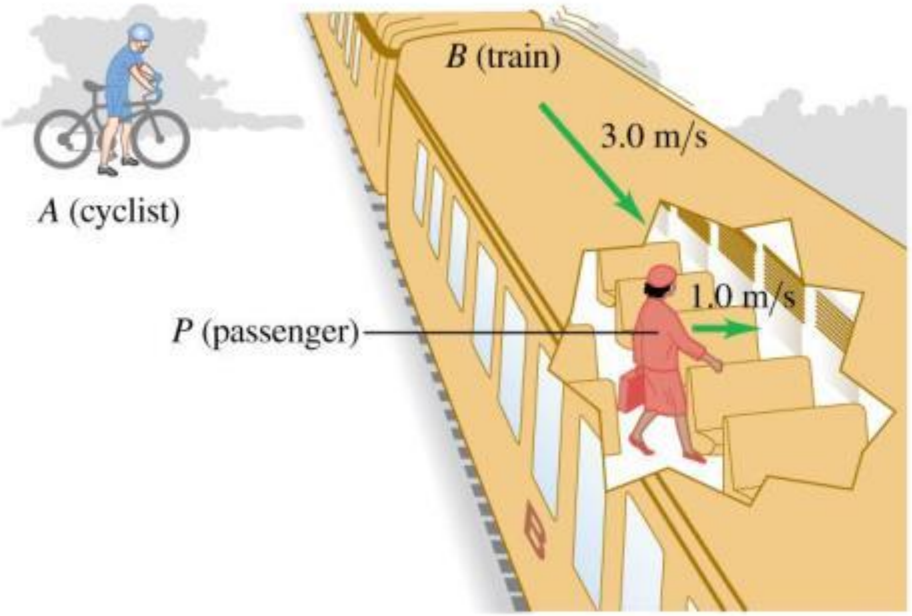


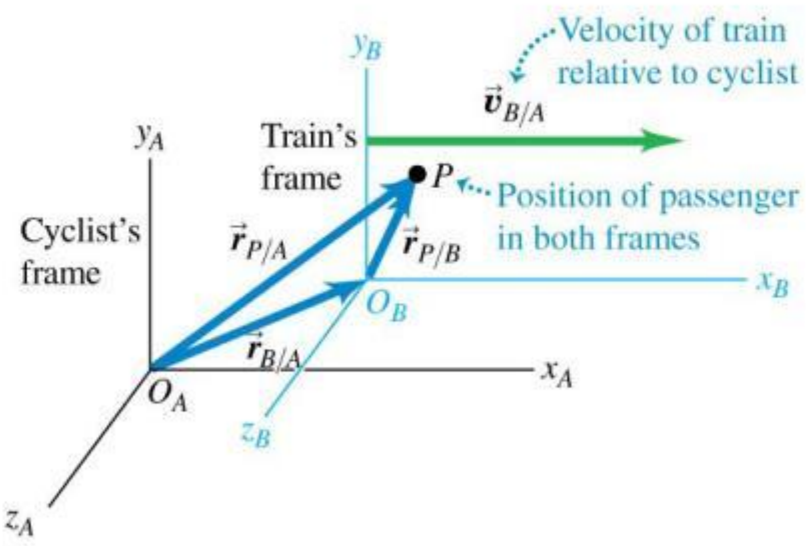
Relative velocity in two or three dimensions

- We extend relative velocity to two or three dimensions by using vector addition to combine velocities.



Relative velocity in two or three dimensions





Relative velocity in space:

$$\vec{v}_{P/A} = \vec{v}_{P/B} + \vec{v}_{B/A}$$

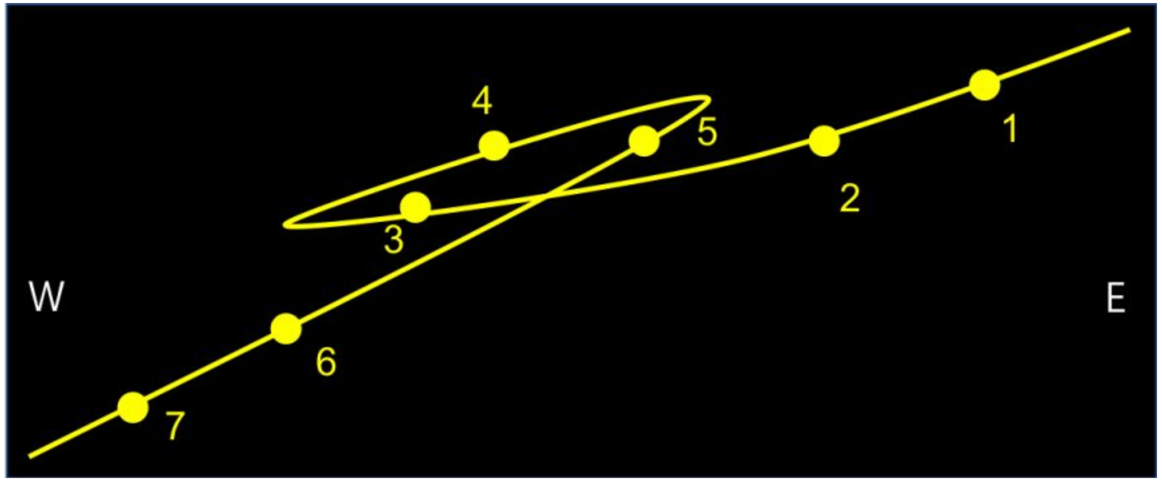
Velocity of P relative to A

Velocity of P relative to B

Velocity of B relative to A

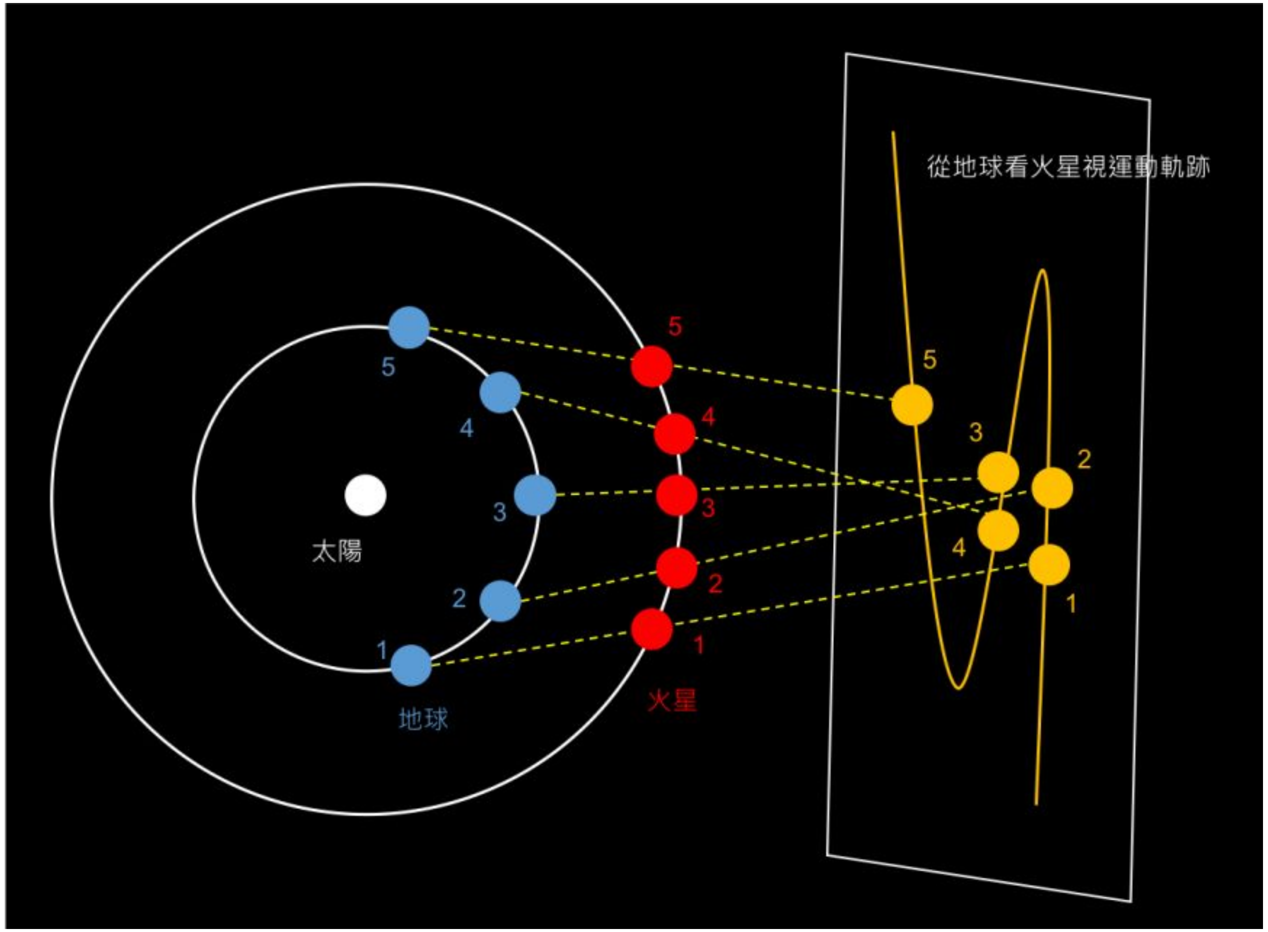
Shui Ni 水逆

- The retrograde motion of Mercury as well as other planets.



Shui Ni 水逆

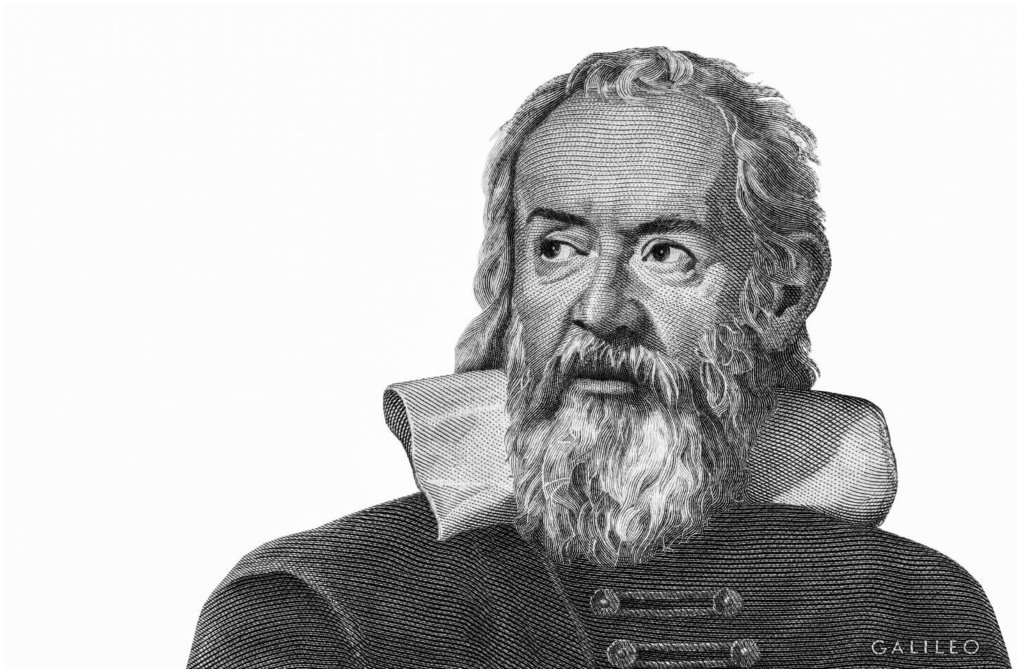
- $\vec{v}_{\text{Mars/Earth}} = \vec{v}_{\text{Mars/Sun}} + \vec{v}_{\text{Sun/Earth}} = \vec{v}_{\text{Mars/Sun}} - \vec{v}_{\text{Earth/Sun}}$



From http://www.geostory.tw/planet_retrograde_heliocentric_model/

Galileo's ship

"Dialogue Concerning the Two Chief World Systems" (1632)



Galileo's ship

- Shut yourself up with some friend in the main cabin below decks on some large ship, and have with you there some flies, butterflies, and other small flying animals. Have a large bowl of water with some fish in it; hang up a bottle that empties drop by drop into a wide vessel beneath it. With the ship standing still, observe carefully how the little animals fly with equal speed to all sides of the cabin. The fish swim indifferently in all directions; the drops fall into the vessel beneath; and, in throwing something to your friend, you need to throw it no more strongly in one direction than another, the distances being equal; jumping with your feet together, you pass equal spaces in every direction.

Galileo's ship

- When you have observed all these things carefully (though doubtless when the ship is standing still everything must happen in this way), have the ship proceed with any speed you like, so long as the motion is uniform and not fluctuating this way and that. You will discover not the least change in all the effects named, nor could you tell from any of them whether the ship was moving or standing still. In jumping, you will pass on the floor the same spaces as before, nor will you make larger jumps toward the stern than toward the prow even though the ship is moving quite rapidly, despite the fact that during the time that you are in the air the floor under you will be going in a direction opposite to your jump.

Galileo's ship

- In throwing something to your companion, you will need no more force to get it to him whether he is in the direction of the bow or the stern, with yourself situated opposite. The droplets will fall as before into the vessel beneath without dropping toward the stern, although while the drops are in the air the ship runs many spans. The fish in their water will swim toward the front of their bowl with no more effort than toward the back, and will go with equal ease to bait placed anywhere around the edges of the bowl. Finally the butterflies and flies will continue their flights indifferently toward every side, nor will it ever happen that they are concentrated toward the stern, as if tired out from keeping up with the course of the ship, from which they will have been separated during long intervals by keeping themselves in the air.

Galileo's ship

- And if smoke is made by burning some incense, it will be seen going up in the form of a little cloud, remaining still and moving no more toward one side than the other. The cause of all these correspondences of effects is the fact that the ship's motion is common to all the things contained in it, and to the air also. That is why I said you should be below decks; for if this took place above in the open air, which would not follow the course of the ship, more or less noticeable differences would be seen in some of the effects noted.

The principle of relativity

- The laws of physics are the same for *inertial frames that are moving with respect to each other*.
- This is the so-called “the principle of relativity” or “Galilean invariance”.

Summary

- 1D motion
- 2D & 3D motion
- Projectile
- Uniform circle
- General curvilinear motion in n-t coordinate
- Relativity